NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Neurodegenerative diseases and Selective Neuropathies

Principles and Practices of Osteopathic Medicine I

Maria M. Plummer, M.D.
Associate Professor
Department of Clinical Specialties
mplummer@nyit.edu

Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives

- 1. Discuss and describe the pathogenesis and morphological (including histological) changes of **Alzheimer disease** and **Pick disease** with clinicopathological correlation.
- 2. Discuss and describe the pathogenesis and morphological (including histologic changes) of **Parkinson disease** with clinicopathological correlation.
- 3. Discuss and describe the pathogenesis and morphological (including histological) changes of **Huntington** disease with clinicopathological correlation.
- 4. Discuss and describe the pathogenesis and morphological (including histological) changes of **ALS** and **prion encephalopathies** with clinicopathological correlation.
- 5. Discuss and describe the pathogenesis and morphological (including histological) changes of selected neuropathies including **Charcot-Marie-Tooth disease, diabetic peripheral neuropathy, and traumatic and compression neuropathy** with clinicopathological correlation.

NEURODEGENERATIVE DISEASES

- Mostly grey matter diseases - **progressive loss of neurons** and associated secondary changes in white matter tracts

- Common theme – **accumulation of protein aggregates (inclusions) resistant to degradation**

- Small protein aggregates – aberrant localization within neurons, elicit stress response in neurons, toxic to neurons; appear as inclusions (micro)

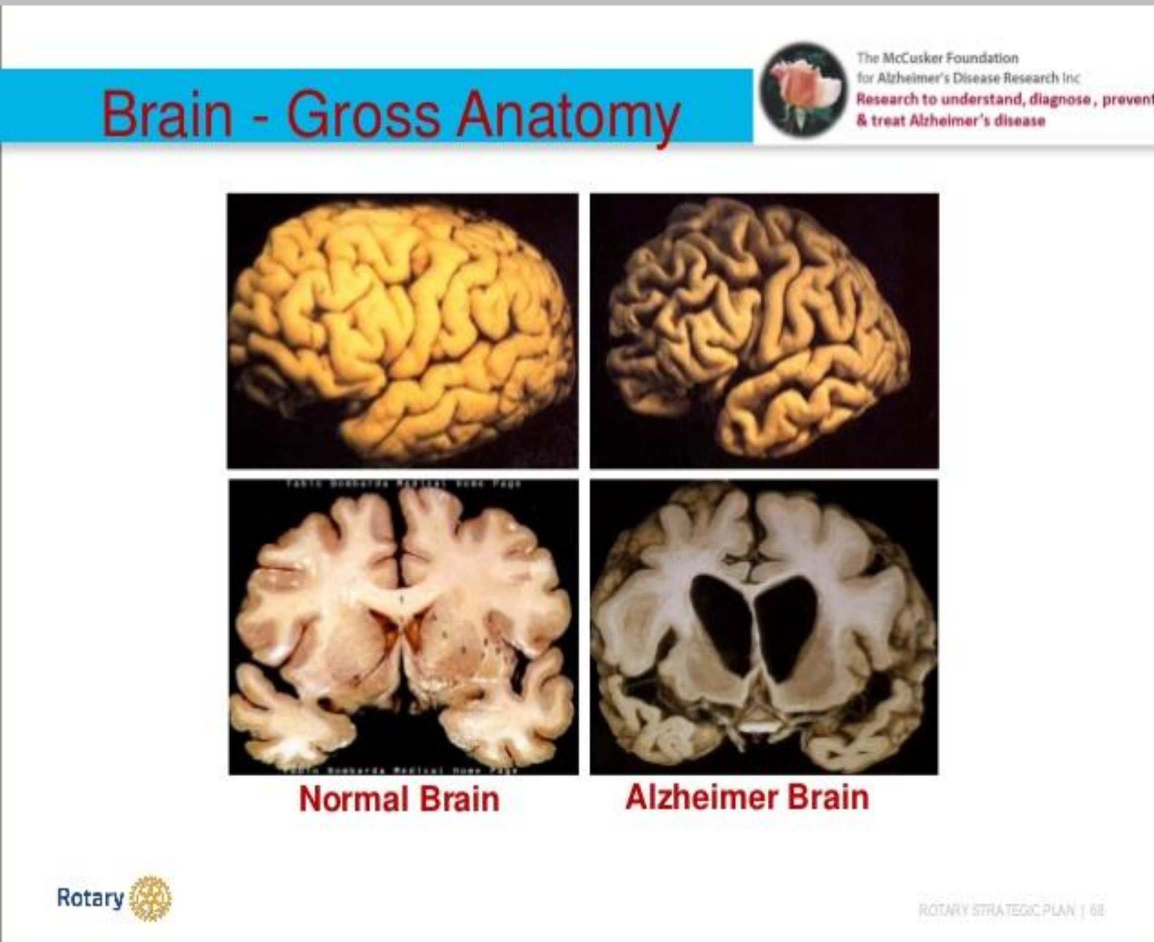
Robbins and Cotran, Pathologic Basis of Disease, 11th ed., 2025, Ch. 28

ALZHEIMER DISEASE

- **Most common cause of dementia in people > 50 years**, increasing incidence with age
- Neurodegenerative disease affecting **cerebral cortex**
- Major clinical manifestation: **dementia;** progressive loss of cognitive function
- Begins as slow impairment of higher cognitive function with changes in mood and behavior; progresses to disorientation, memory loss, visuospatial disorientation, judgement problems, personality changes and aphasia over 5 – 10 year period or longer
- Patients become severely disabled, mute, immobile; pneumonia usually terminal event
- **Most cases sporadic; 5 – 10% cases familial**
- **Pathological examination of brain tissue at autopsy needed for definitive diagnosis** though clinical assessment and modern radiology can make premortem diagnosis in 80 to 90% of cases

Robbins and Cotran, Pathologic Basis of Disease, 11th ed., 2025, Ch. 28

Alzheimer's disease



- **Cortical atrophy** with gyral narrowing and sulcal widening especially in frontal, temporal, and parietal lobes
- With significant atrophy - compensatory ventricular enlargement **(hydrocephalus ex vacuo)** secondary to reduced brain volume

https://image.slidesharecdn.com/copingandcaringwithalzheimersdisease-150608185522-lva1-app6891/95/coping-and-caring-with-alzheimers-disease-and-dementia-your-action-group-in-action-68-638.jpg?cb=1433789954

Robbins and Cotran, Pathologic Basis of Disease, 10th ed., 2020, Ch. 28

AD - pathogenesis

- **Accumulation of two proteins (Aβ and tau)** in specific brain regions, likely because of excessive production and defective removal
- Two pathologic hallmarks of AD: **amyloid (neuritic) plaques and neurofibrillary tangles**
- **Plaques:** deposits of **aggregated Aβ (A-beta) peptides** in the neuropil
- **Tangles**: **aggregates of microtubule binding protein tau**; develop intracellularly and persist extracellularly after neuronal death
- Both plaques and tangles contribute to neural dysfunction
- **Number of neurofibrillary tangles correlates better with the degree of dementia than does the number of neuritic plaques**
- Progressive, eventually severe, neuronal loss and reactive gliosis in same regions of plaques and tangles

Robbins and Cotran, Pathologic Basis of Disease, 11th ed., 2025, Ch. 28

Role of A-beta in AD

- Fundamental problem **: deposition of A-beta peptides, derived from processing of APP (amyloid precursor protein)**
- **Aβ generation seems to be critical initiating event for development of AD**
- Processing of APP begins with cleavage in extracellular domain, followed by intramembranous cleavage
- Two extracellular sites of cleavage – may be by two different proteases, α-secretase and β-secretase
- IF first cut occurs at α-secretase site: Aβ not generated (non-amyloidogenic pathway)
- Surface APP can be endocytosed into vesicles, where it can be cleaved by β-secretase (cuts at slightly more N-terminal site within APP: the **amyloidogenic pathway**)

Robbins and Cotran, Pathologic Basis of Disease, 11th ed., 2025, Ch. 28

Role of A-beta in AD (continued)

- Following cleavage of APP at either of these sites, the γ-secretase complex (contains presenilin-1 or presenilin-2, nicastrin, PEN2, and APH1) performs an intramembranous cleavage
- **Cleavage by α-secretase and β-secretase releases Aβ42, capable of aggregation and amyloid formation**
- Aβ42 highly prone to aggregation—first into small oligomers (toxic form responsible for neuronal dysfunction); eventually into large aggregates and fibrils visualized by microscopy

Robbins and Cotran, Pathologic Basis of Disease, 10th ed., 2020, Ch. 28

AD, genetics

- Familial forms of AD support Aβ generation as critical step for initiation of AD pathogenesis
  - **Gene encoding APP, on chromosome 21** (in Down syndrome region) – Down syndrome patients show cognitive impairment and histopathologic AD findings in 2nd and 3rd decades of life followed by neuro decline 20 years later
  - Point mutations in APP are another cause of **familial AD**
  - Chrom. 14; gene for presenilin-1 leads to increased A-beta production, especially A-beta 42, and **early onset familial AD**
  - Chrom.1; gene for presenilin-2 also leads to increased A-beta production, especially A-beta 42 and **early onset familial AD**
  - Chrom. 19: gene for apolipoprotein E leads to mutation in allele epsilon 4 resulting in increase risk of development of AD and **decrease age at onset of AD** (**ApoE4 – increased risk**) – patients with epsilon 4 allele overrepresented in AD

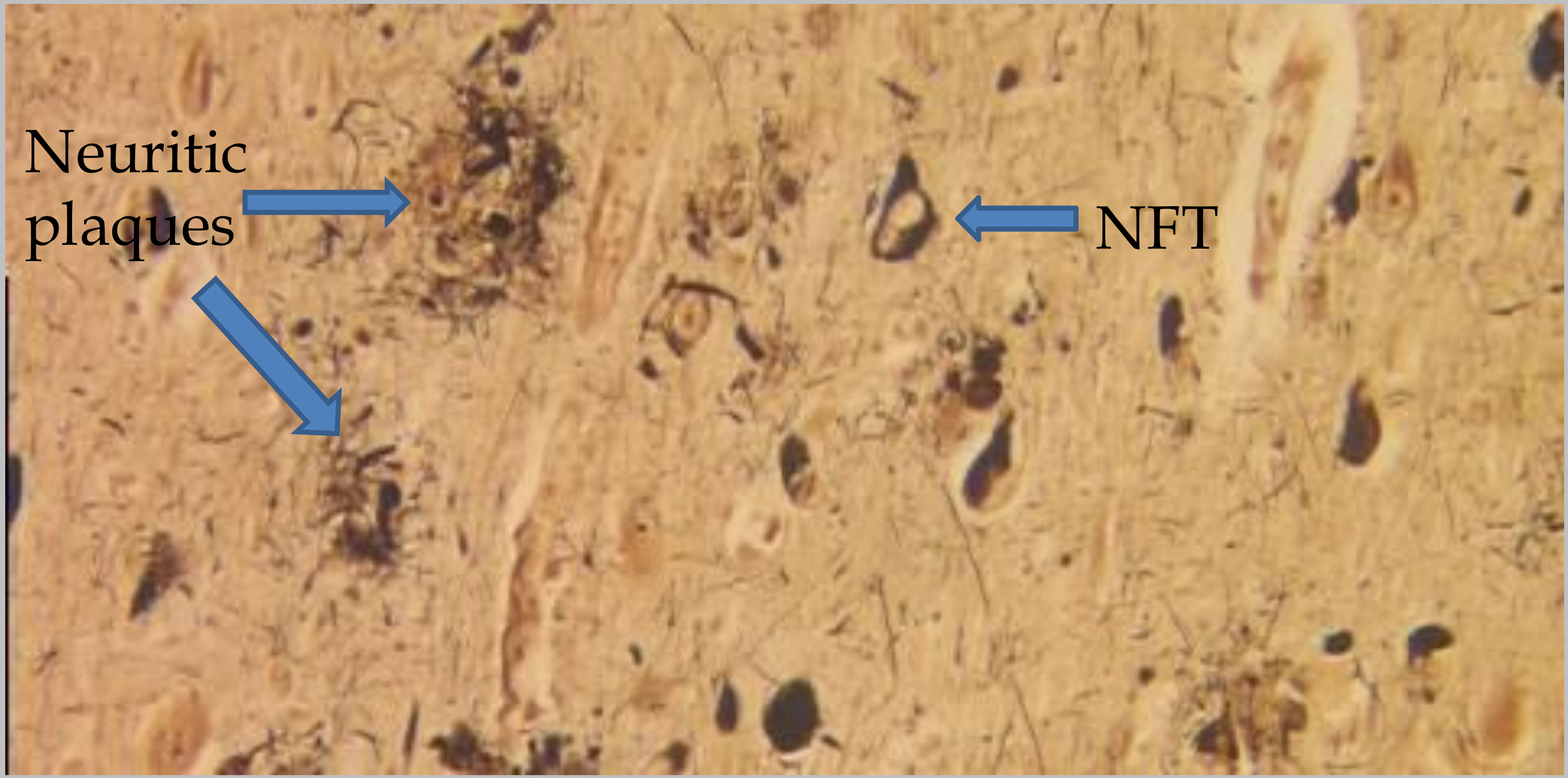
Robbins and Cotrin, Pathologic Basis of Disease, 10th ed., 2020, Ch. 28

Role of tau protein and inflammation in AD

- **Tau protein**
  - **Microtubule-associated protein present in axons in association with microtubular network**
  - **Development of tangles in AD, becomes hyperphosphorylated, - loses ability to bind to microtubules**
  - Tangles may elicit a stress response or loss of tau may destabilize microtubules

- **Inflammation**
  - Small aggregates and larger deposits of Aβ elicit inflammatory response from microglia and astrocytes
  - Probably helps in clearance of aggregated peptide, but may also stimulate secretion of mediators causing damage

Robbins and Cotrin, Pathologic Basis of Disease, 11th ed., 2025, Ch. 28



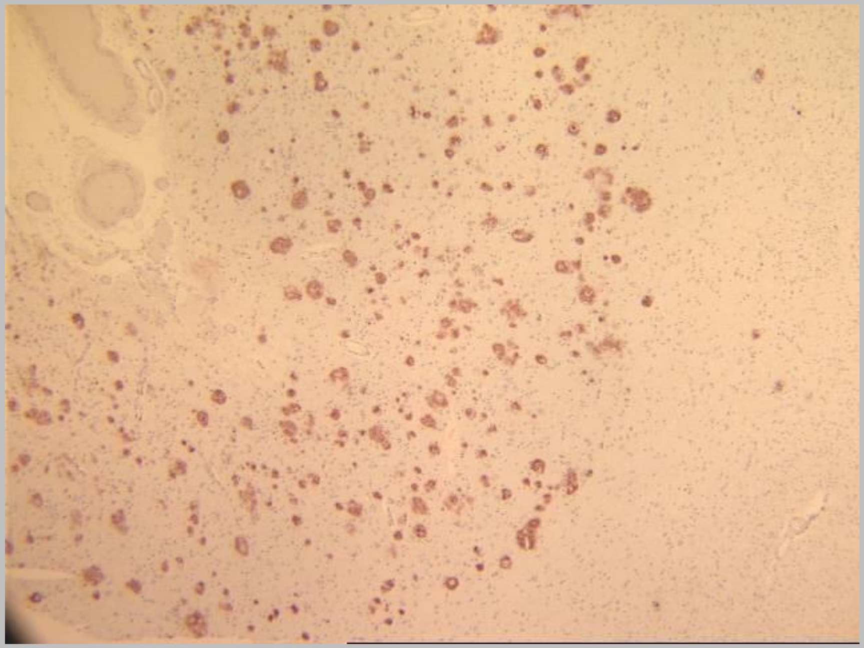
Neuritic plaques – focal, spherical collections of dilated, tortuous, neuritic processes usually around a central amyloid core, surrounded by clear halo; found in hippocampus, amygdala, and neocortex, with relative sparing of primary motor and sensory cortices (also for neurofibrillary tangles)

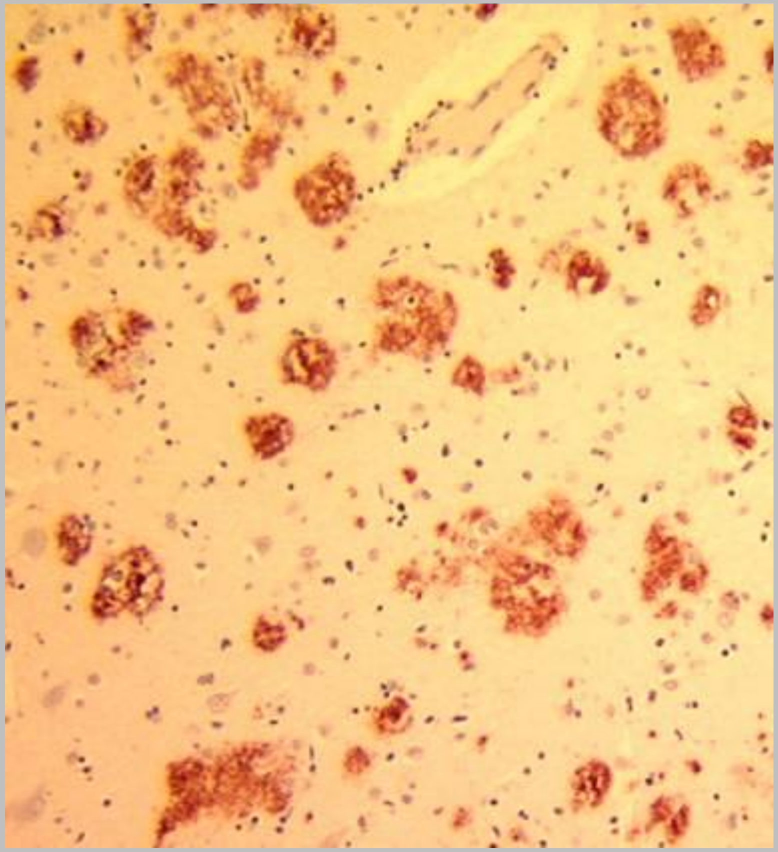
(NFT = neurofibrillary tangle)

Robbins and Cotran, Pathologic Basis of Disease, 10th ed., 2020, Ch. 28

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

AD: amyloid core (Congo red) – mainly made up of **A-beta peptide,** processed from proteolytic cleavage of amyloid precursor protein, APP

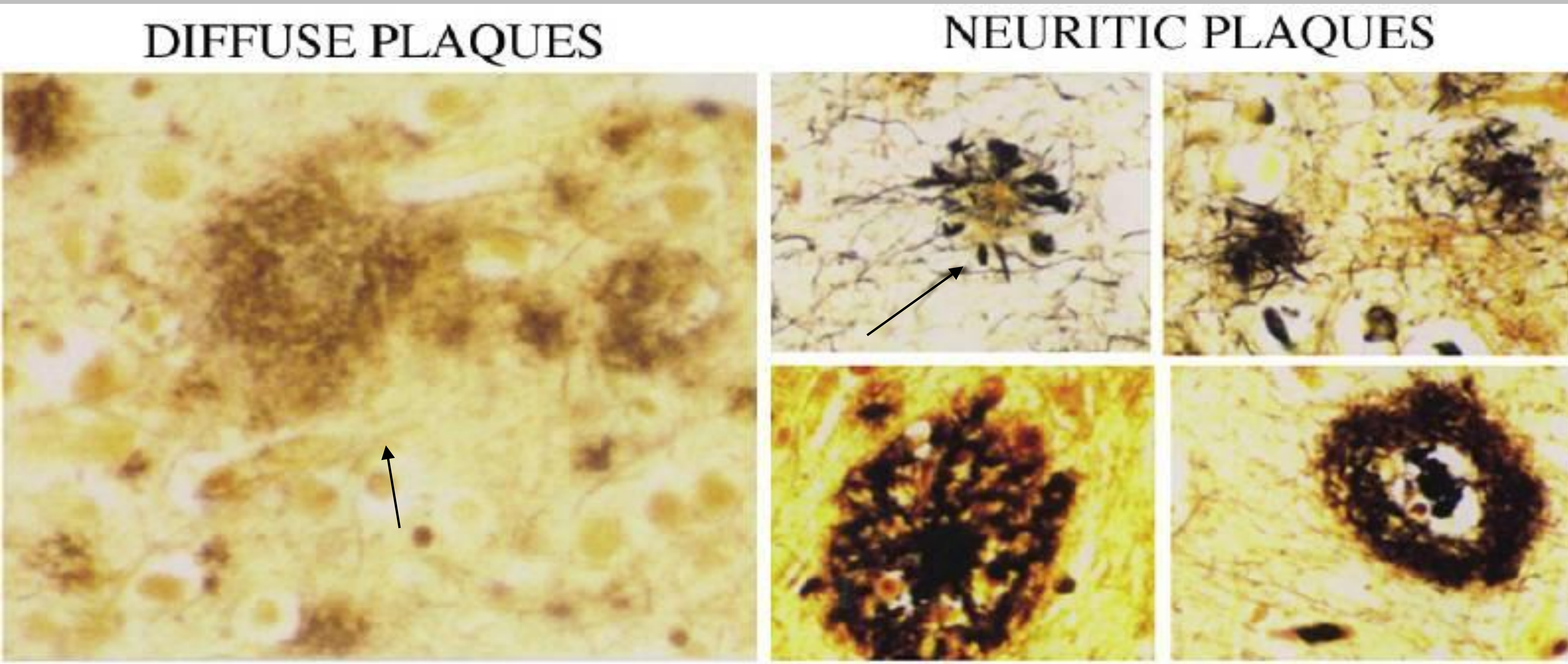




NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Robbins and Cotran, Pathologic Basis of Disease, 10th ed., 2020, Ch. 28

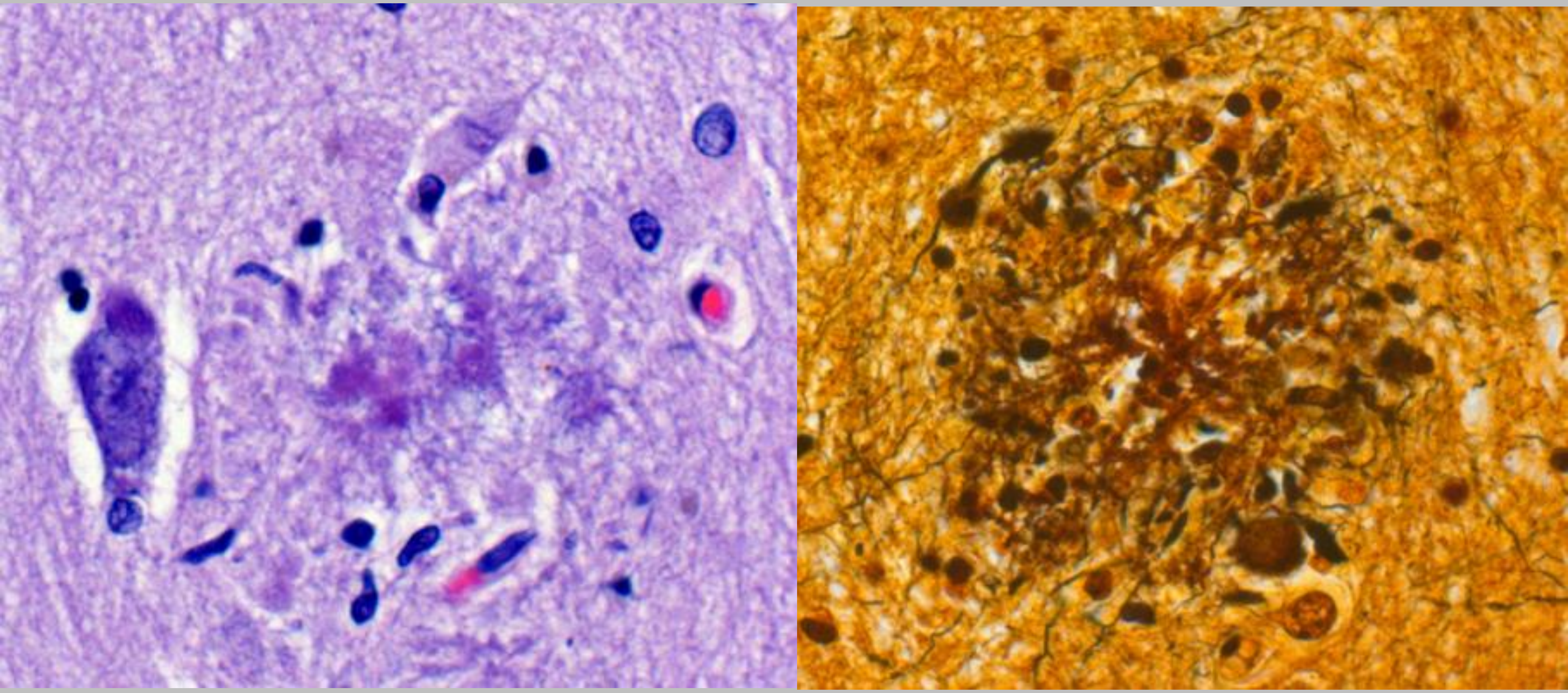
Plaques: diffuse or neuritic



- “Senile” plaques seen with Bielschowsky have been divided into diffuse and neuritic subtypes. from CERAD handbook
- **Diffuse** – no surrounding neuritic reaction, probably represents **early stage plaque (mostly A-beta 42)** mostly in superficial parts of cerebral cortex, basal ganglia, cerebellar cortex
- **Neuritic** plaques contain both **Aβ 40 and Aβ 42** .

Robbins and Cotran, Pathologic Basis of Disease, 10th ed., 2020, Ch. 28

Neuritic plaques : H and E v. Silver stain

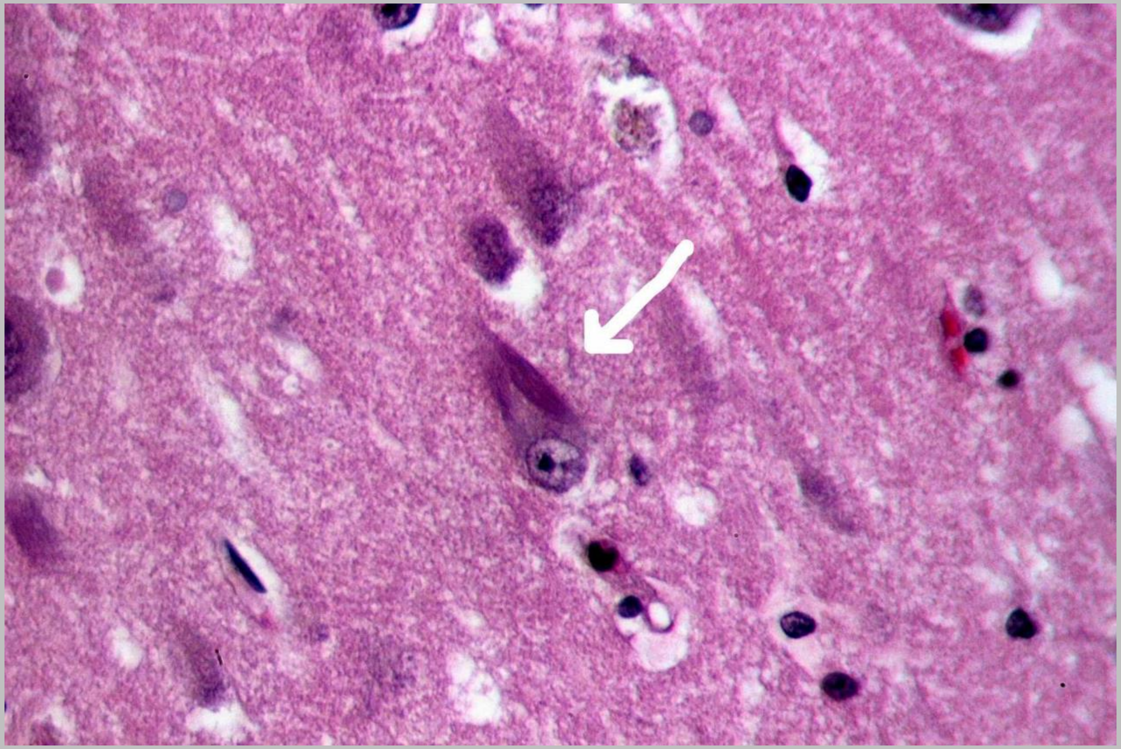


Neurofibrillary tangles

NFTs- tau containg bundles of filaments in cytoplasm of neurons which displace or encircle nucleus

Commonly found in cortical neurons, especially in entorhinal cortex, as well as in pyramidal cells of hippocampus, amygdala, basal forebrain, and raphe nuclei

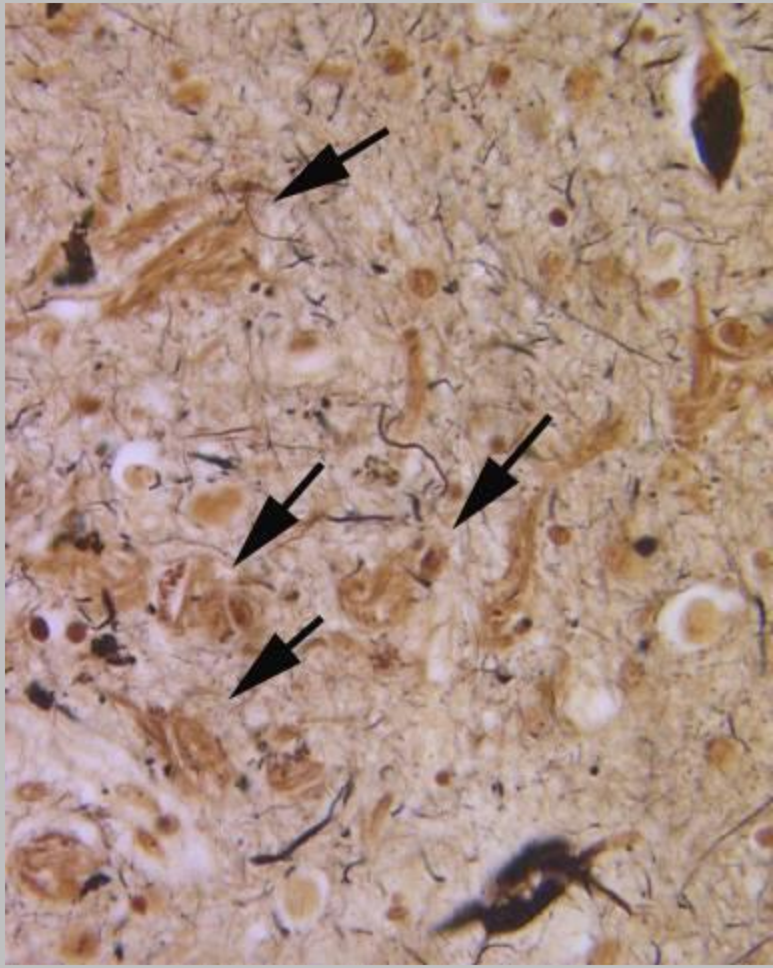
Unlike neuritic and diffuse plaques, **NFTS found in other neurodegenerative - not specific for AD**



https://www.cureus.com/articles/27530-the-neuroprotective-effects-of-exercise-on-cognitive-decline-a-preventive-approach-to-alzheimer-disease

Robbins and Cotran, Pathologic Basis of Disease, 10th ed., 2020, Ch. 28

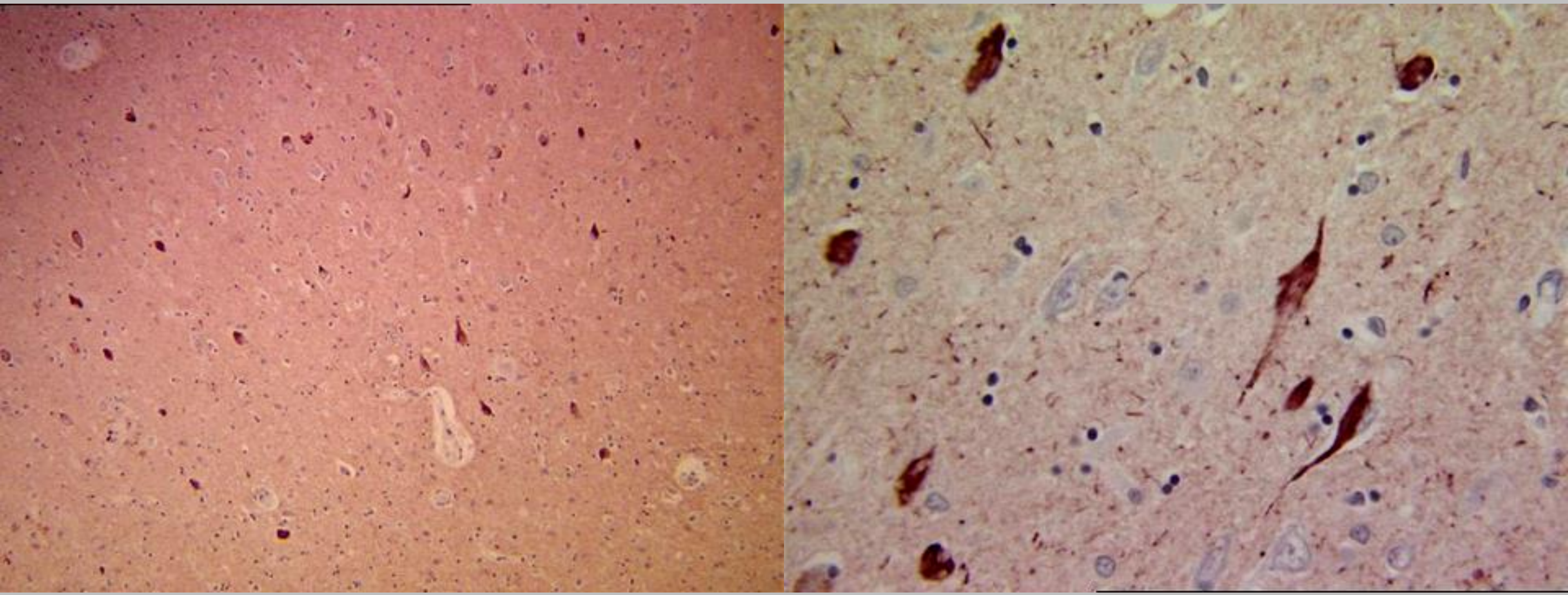
Neurofibrillary tangles



- NFTs composed of fibrillar aggregates made of paired helical filaments
- Insoluble, resitant to clearance in vivo, remaining visible as "ghost or tombstone tangles" (right, arrows) after death of host neuron
- **The number of NFTs correlates better with the degree of dementia than does the number of neuritic plaques**

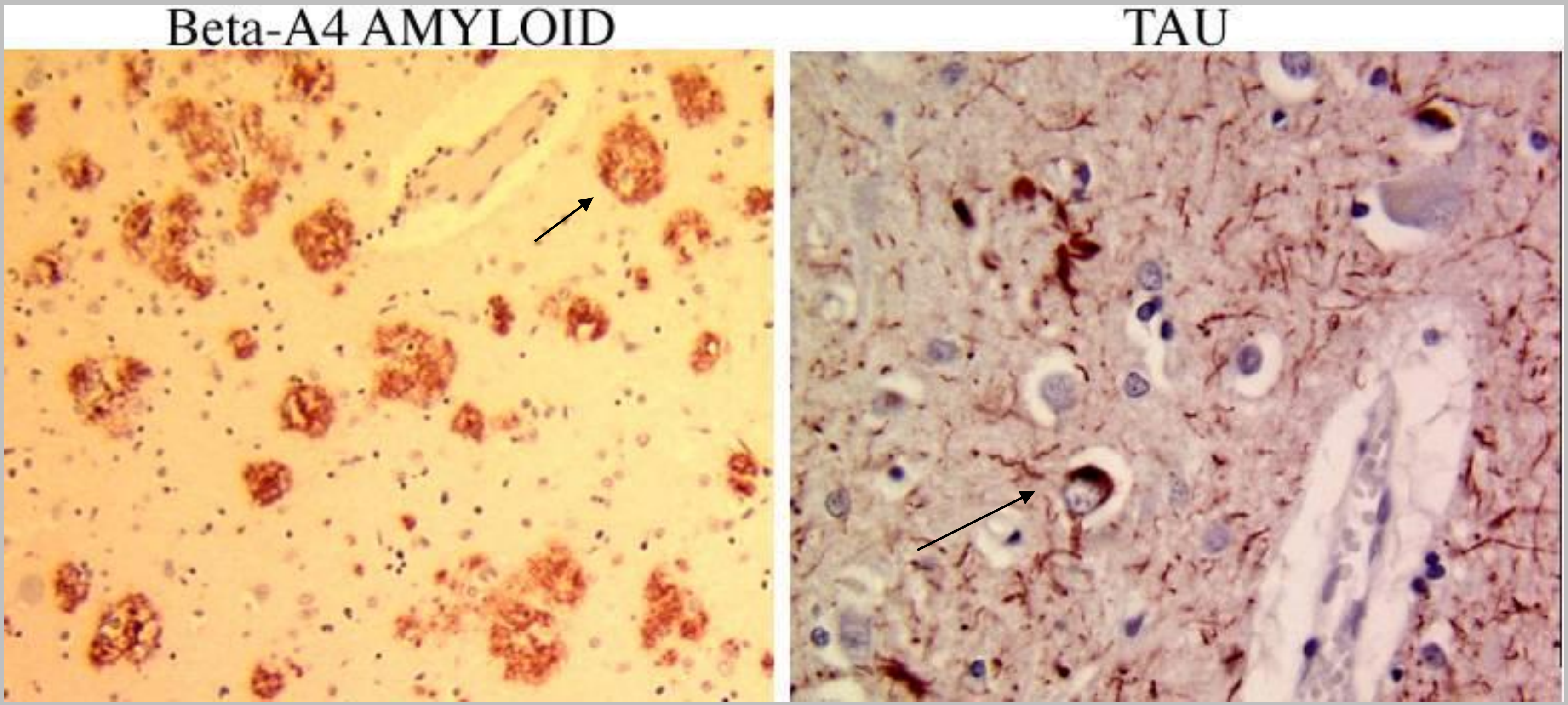
Robbins and Cotran, Pathologic Basis of Disease, 10th ed., 2020, Ch. 28

Tau accumulation in AD



- NFTs are made of abnormal accumulations of hyerphosphorylated tau protein.
- In late stages of AD, NFTs are frequent in neocortical areas and accompanied by numerous neuropil threads (tau immunostains above)

AD: Amyloid and Tau



- AD characterized by extracellular accumulation of Abeta amyloid (plaques) and intracellular accumulation of tau (neurofibrillary tangles and neuropil threads)

Robbins and Cotran, Pathologic Basis of Disease, 10th ed., 2020, Ch. 28.

AD Biomarkers

- Biochemical markers correlated with degree of dementia:
  - loss of choline acetyltransferase
  - loss of synaptophysin immunoreactivity
  - amyloid burden
- Aβ deposition in brain may be shown using imaging methods (with18F-labeled amyloid-binding compounds): can identify **asymptomatic patients who in early stages of AD**
- **AD pathology also shows increased phosphorylated tau and reduced Aβ in the CSF -** biomarkers helpful in identification of preclinical stages of AD, before dementia
- Pharm clinical trials more shifted to prevention of earlier disease, e.g., clearance of A-beta via immunological approaches or disruption of generation of A-beta with agents that target secretases, as well as prevention of alterations in tau

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Robbins and Cotran, Pathologic Basis of Disease, 10th ed., 2020. Ch. 28

PICK DISEASE – Frontotemporal lobar degeneration - tau

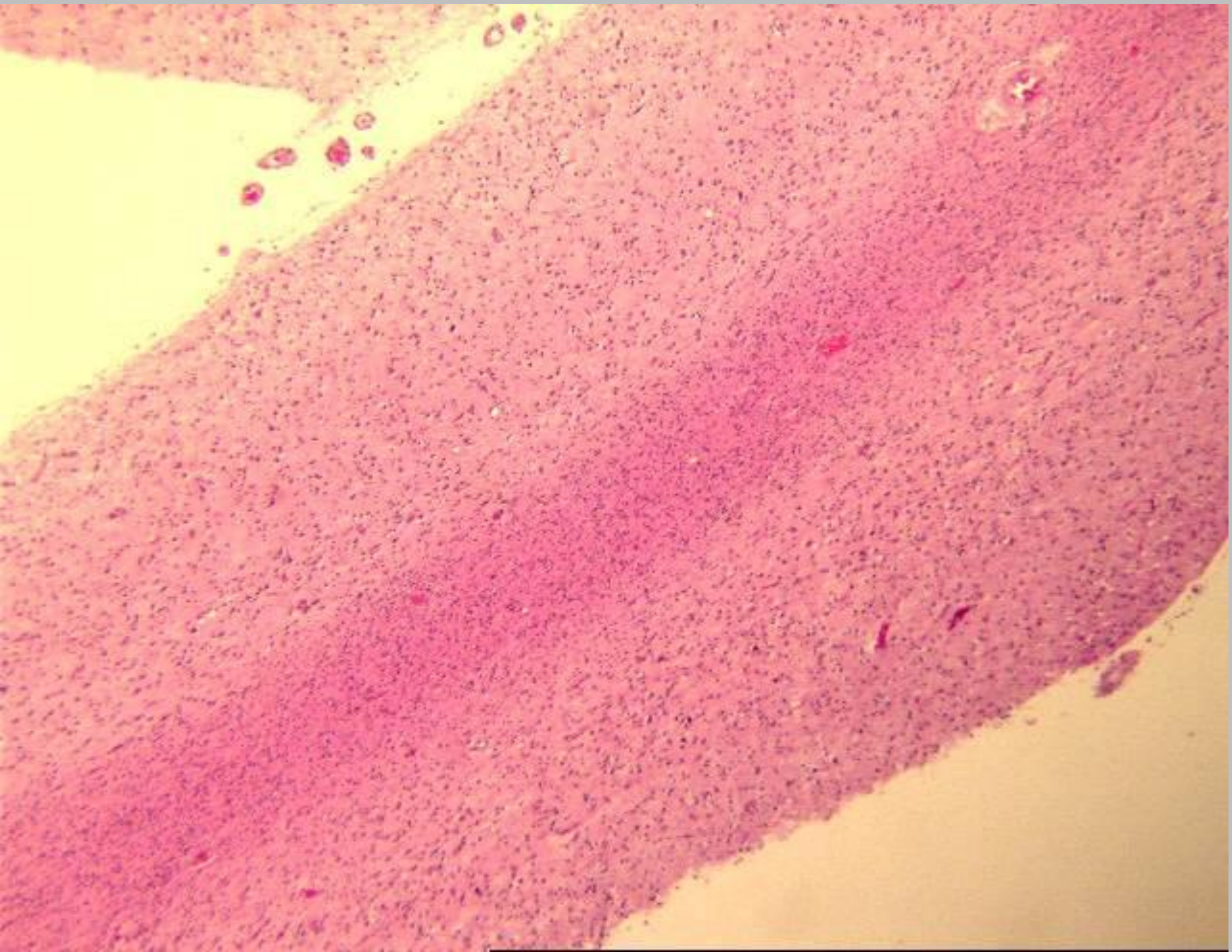
- FTLDs - heterogeneous group of disorders with focal degeneration of frontal and/or temporal lobes; clinically distinguished from AD: alterations in personality, behavior, and language (aphasias) **preceding** memory loss; postmortem pathologic evaluation required for definitive diagnosis
- FTLDs associated with cellular inclusions of specific proteins - two most common patterns: tau inclusions (FTLD-tau) and TDP-43 inclusions (FTLD-TDP)
- FTLD – tau: forms of FTLD where affected cortical regions show progressive neuronal loss and reactive gliosis, and cytoplasmic tau inclusions in neurons
- FTLD-tau may be associated with mutations in MAPT, gene that encodes tau, or may arise sporadically
- Inverse relationship between degree of tau phosphorylation and ability to bind microtubules; Phosphorylated tau tends to aggregate
- **Pick disease** (subtype of FTLD-tau) - brain shows pronounced, usually asymmetric **atrophy of frontal and temporal lobes,** sparing of posterior two-thirds of the superior temporal gyrus

Robbins and Cotran, Pathologic Basis of Disease, 10th ed. 2020, Ch. 28.

Pick Disease: severe atrophy of the frontal and temporal lobes (lobar atrophy), often with “knife-edge” atrophy of the gyri. Thin, wafer-like gyri.

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Pick Disease



Severe neuronal loss with intense gliosis in affected cortical regions

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Case Question

An autopsy study is performed of patients who died from 55 to 80 years of age. A subset of these patients had cerebral atrophy and brain weights that were less than normal for age and body size. The brains showed hydrocephalus ex vacuo, but no focal lesions. Microscopic examination showed the presence of increased numbers of neurofibrillary tangles and neuritic plaques within the cerebral cortex. Which of the following was most likely present in those groups of patients?

A. symmetric muscle weakness
B. gait disturbances
C. grand mal seizures
D. progressive dementia
E. choreiform movements

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Parkinson Disease

- **Neurodegenerative hypokinetic movement disorder caused by loss of dopaminergic neurons from substantia nigra**
- Clinical - diminished facial expression (masked facies), stooped posture, slowing of voluntary movement, festinating gait (shuffling), rigidity, and "pill-rolling" tremor: (TRAPS – tremor, rigidity, akinesia (or bradykinesia), postural instability, shuffling gait)

- Similar symptoms can occur with dopaminergic antagonists or toxins that selectively damage the dopaminergic system
- Clinical triad— **tremor, rigidity, and bradykinesia**—in absence of toxic or other known underlying etiology - confirmed by symptomatic response to L-DOPA replacement therapy

Robbins and Cotran, Pathologic Basis of Disease, 11th ed. 2025, Ch. 28

Parkinson disease

- **Degeneration of substantia nigra** (motor symptoms) represents mid-stage in progression from involvement lower in brainstem and eventually cerebral cortex, leading to cognitive impairment
- Note: acute parkinsonian syndrome and destruction of neurons in substantia nigra follows exposure **to MPTP** (1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine), contaminant in synthesized illegal batches of opioid meperidine
- Pesticide exposure may be risk factor for PD; caffeine and nicotine appear to be protective
- Dopaminergic neurons of substantia nigra project to the striatum; **degeneration in PD associated with reduction in the striatal dopamine content**
- **Severity of motor syndrome proportional to dopamine deficiency**
- Can be partially corrected by L-Dopa (immediate precursor of dopamine); does not reverse morphological changes or arrest progress of disease
- As disease progresses, drug therapy less effective

Robbins and Cotran, Pathologic Basis of Disease, 10th ed, 2020, Ch. 28

Pathogenesis of PD

Protein accumulation and aggregation, mitochondrial abnormalities, and neuronal loss in the substantia nigra and other areas of brain

Most PD is sporadic, but some **genetic causes** help explain pathogenesis:

•First mutations identified in **autosomal dominant PD involve SNCA**, gene that encodes **α-synuclein**, lipid-binding protein normally localized to synapses; major component of the **Lewy body, diagnostic hallmark of PD**

•**α-synuclein forms aggregates**; toxic to neurons

OF INTEREST:

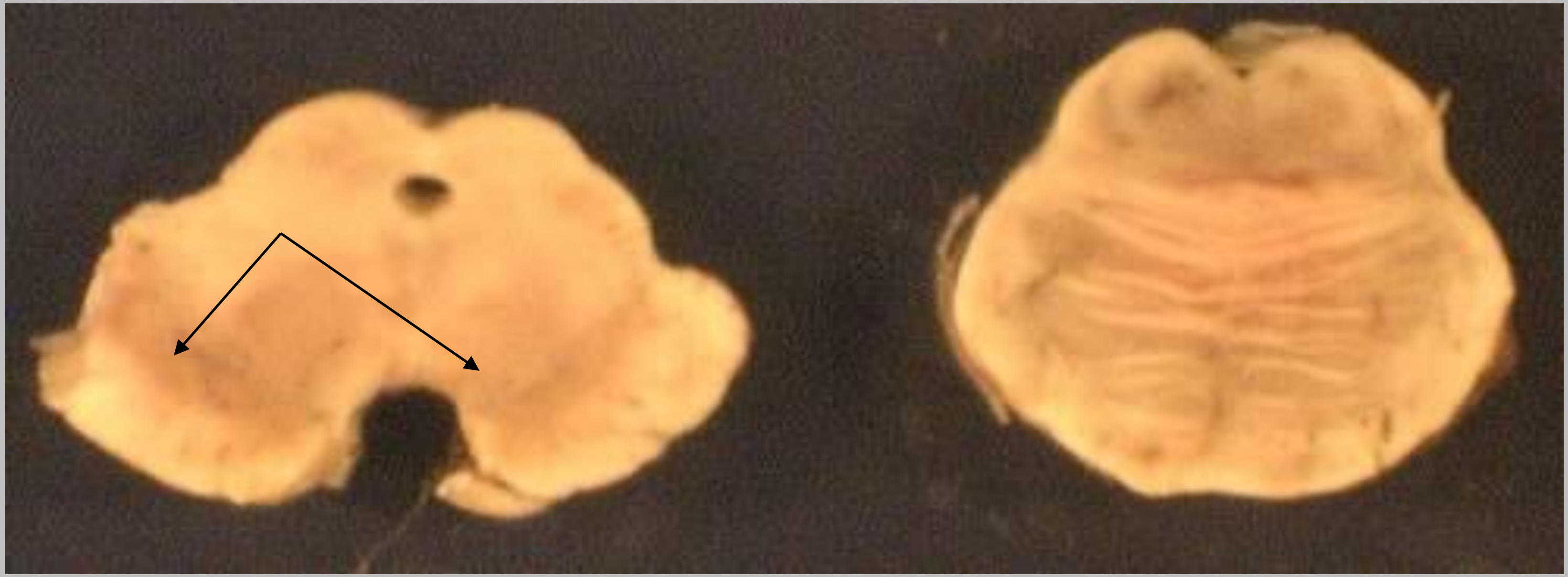
•*Mitochondrial dysfunction implicated as contributing factor based on autosomal recessive forms of PD caused by mutations in genes that encode the proteins DJ-1, PINK1, and parkin*

•*In about 5% of PD: Heterozygous mutations in lysosomal enzyme glucorebrosidase most important risk factor for development of PD- suggests dysfunction of lysosome-autophagy pathway contributing to PD pathogenesis, most likely via abnormal metabolism of α-synuclein, activation of unfolded protein response, and increased inflammation*

•*Mutations in gene encoding LRRK2 (leucine-rich repeat kinase 2) most common cause of autosomal dominant PD and also found in some late-onset, apparently sporadic cases*

Robbins and Cotran, Pathologic Basis of Disease, 10th ed, 2020, Ch. 28

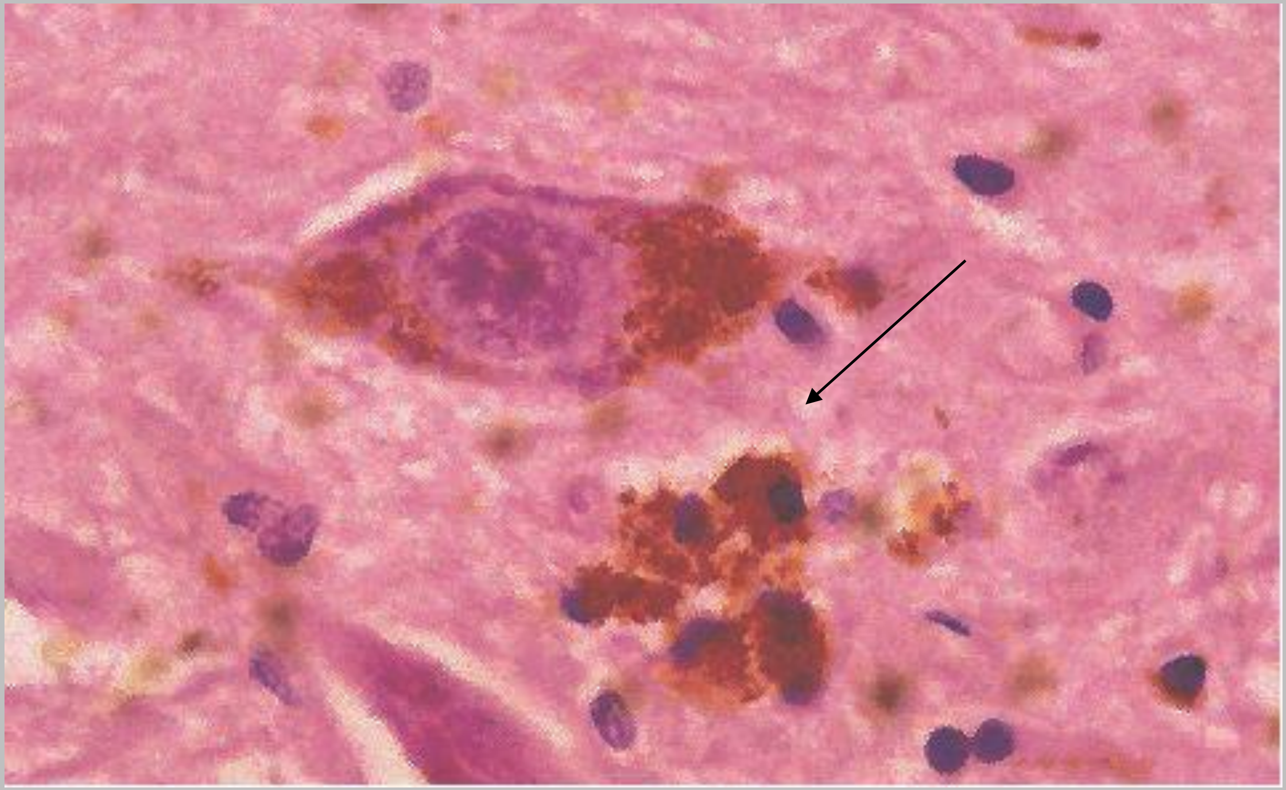
Parkinson's Disease



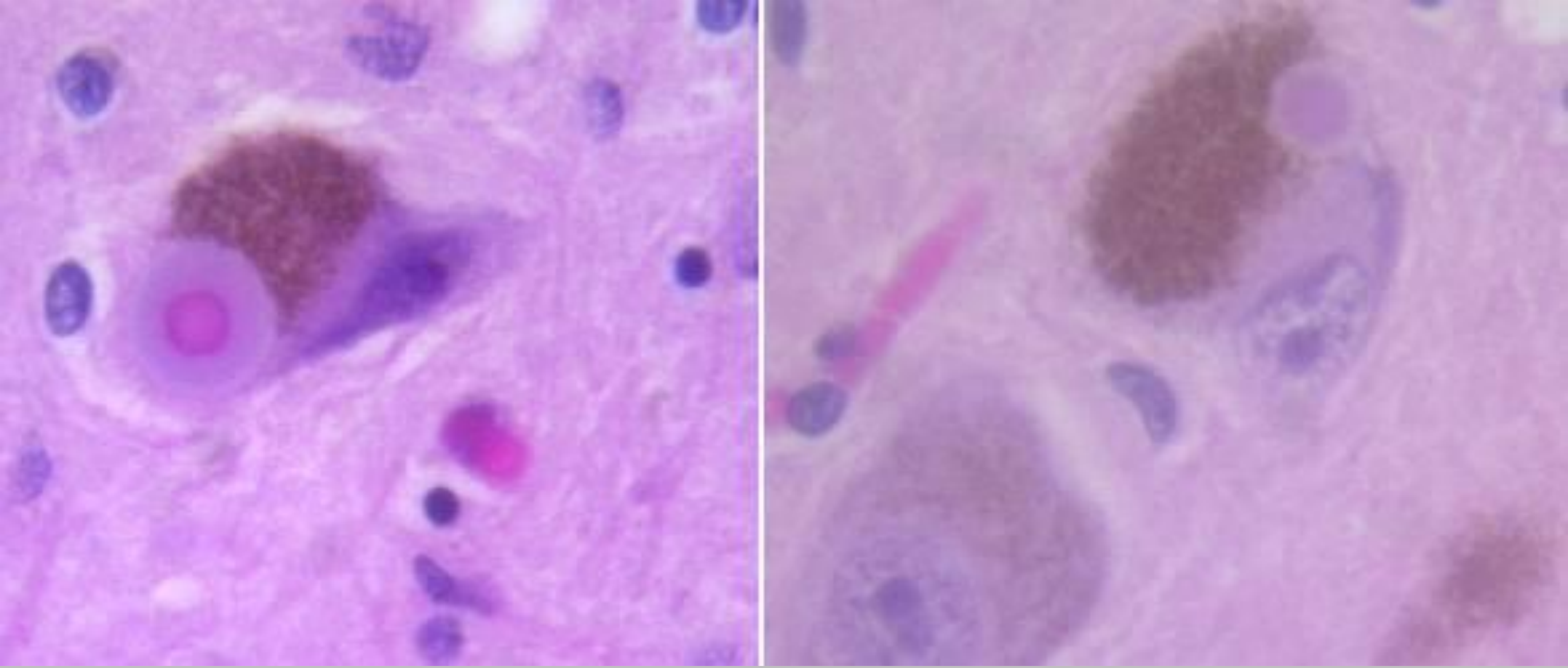
Loss of pigment (dopa catecholaminergic neurons) in the substantia nigra and locus ceruleus - pallor

Robbins and Cotran, Pathologic Basis of Disease, 9th ed.

Parkinson’s Disease



Neuronal loss in the substantia nigra accompanied by numerous pigment-laden macrophages (melanophages, arrow) and gliosis

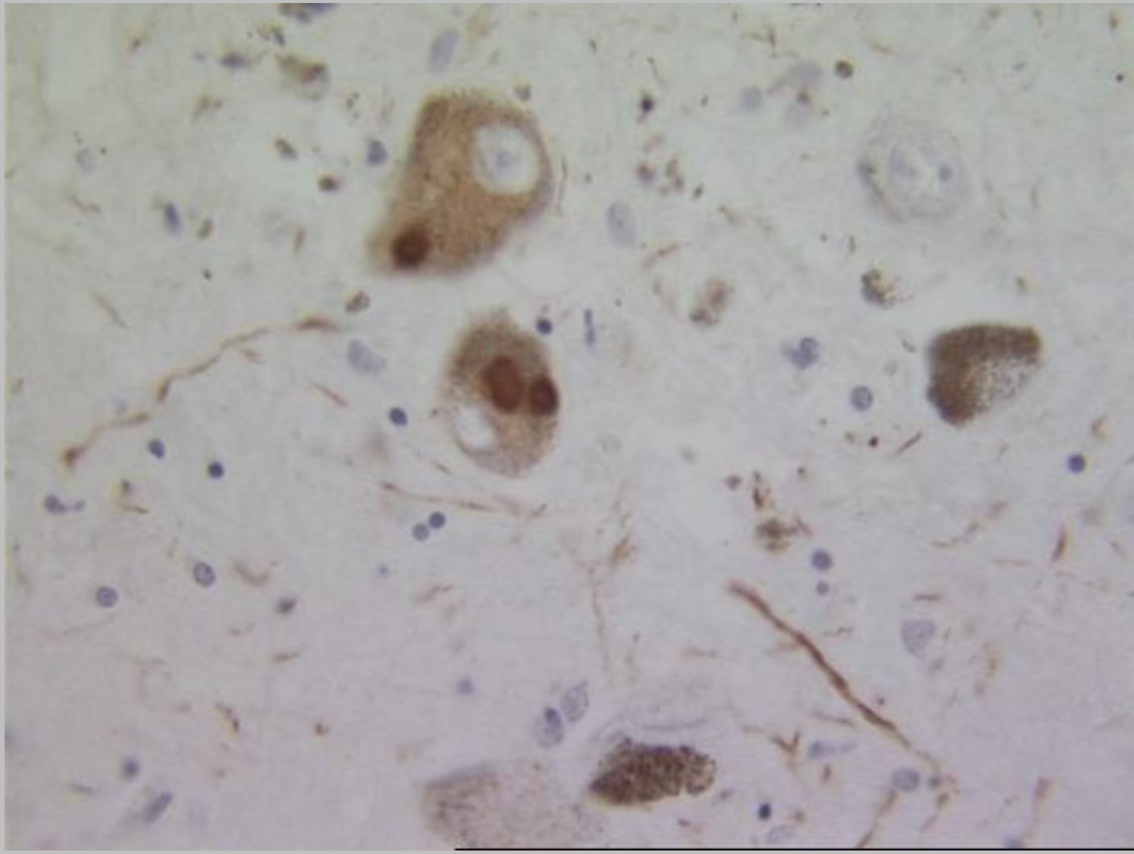


Lewy bodies in substantia nigra – cytoplasmic, eosinophilic, round/elongated inclusions with dense core surrounded by pale halo – EM shows fine filaments, densely packed in core but loose at rim; composed of α-synuclein

Roins and Cotran, Pathologic Basis of Disease, 10th ed., 2020, Ch. 28.

Lewy Bodies

- PD associated with alpha-synuclein (lipid binding protein associated with synapses)
- Lewy bodies: abnormal aggregates of alpha-synuclein.
- Immunostaining for alpha-synuclein strongly labels both Lewy Bodies and Lewy neurites (dystrophic processes containing aggregated α-synuclein) in substantia nigra.



Robbins and Cotran, Pathologic Basis of Disease, 10th ed, 2020, Ch. 28

Case Question

A 70 year old man has had increasing difficulty with voluntary movements because of muscular rigidity. On physical exam, he has difficulty initiating movement, but he can keep moving if he follows someone walking ahead of him. He has an expressionless facies. When he is sitting, his hands have a resting, "pill-rolling" tremor. Which of the following histopathological findings is most likely to be present in his brain?

A. Cerebral neurofibrillary tangles and neuritic plaques
B. Neuronal loss with gliosis in the caudate and putamen
C. Loss of pigmented neurons in the substantia nigra
D. Hemosiderosis and gliosis in the mamillary bodies
E. Acute inflammation and bacteria in the meninges

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Huntington Disease

- Autosomal dominant, progressive movement disorder with dementia caused by degeneration of striatal neurons– progressive and fatal; avg. course of disease 15 years
- **Jerky, hyperkinetic, sometimes dystonic movements involving all parts of body (chorea)** - later may develop bradykinesia and rigidity

- Chorea:

  Sudden jerky, irregular movements with muscle contractions
- Athetosis:

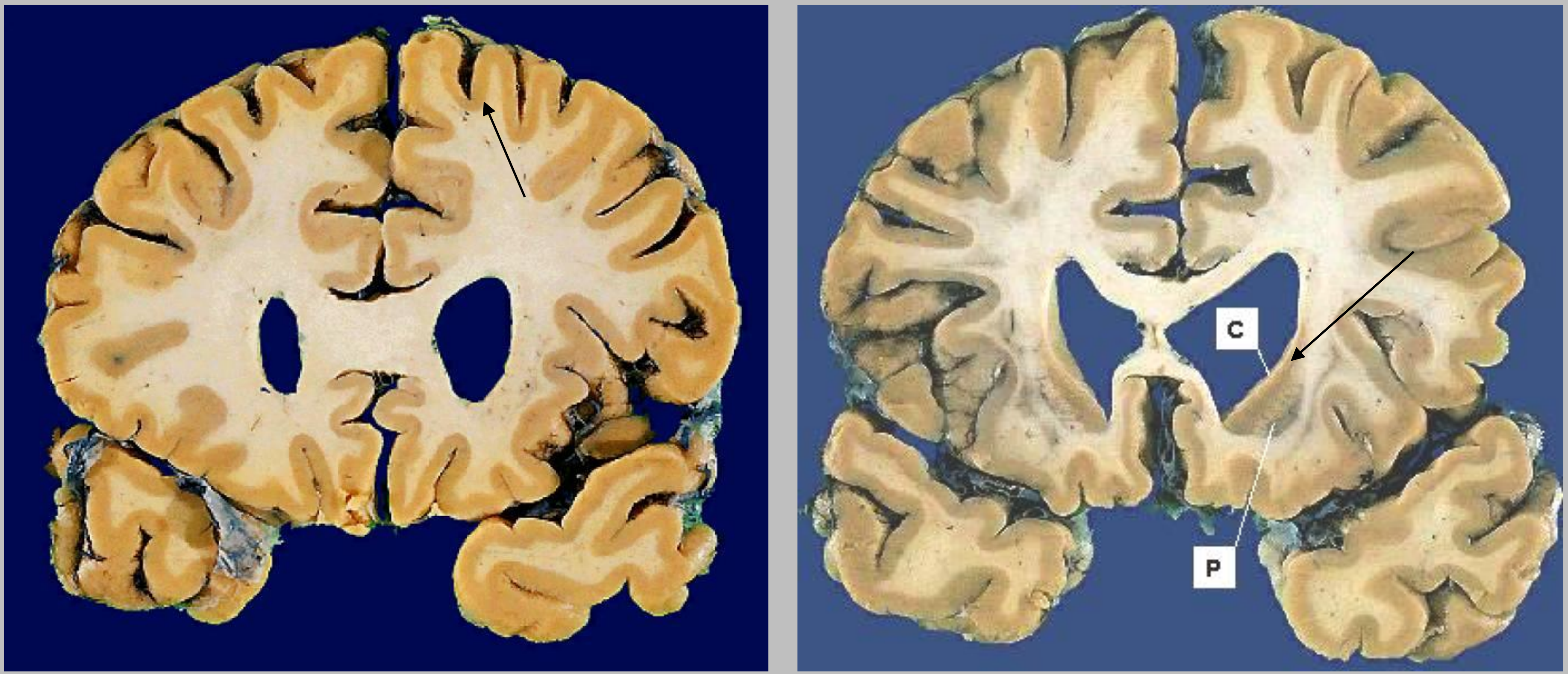
  Twisting and writhing motions
- Dystonia:

  Sustained muscle contractions - usually result of repetitive motions

Summary: Progressive chorea
Hypertonicity (Fidgety movements)
Behavioral disturbances
Dementia

Robbins and Cotran, Pathologic Basis of Disease, 11th ed, 2025, Ch. 28

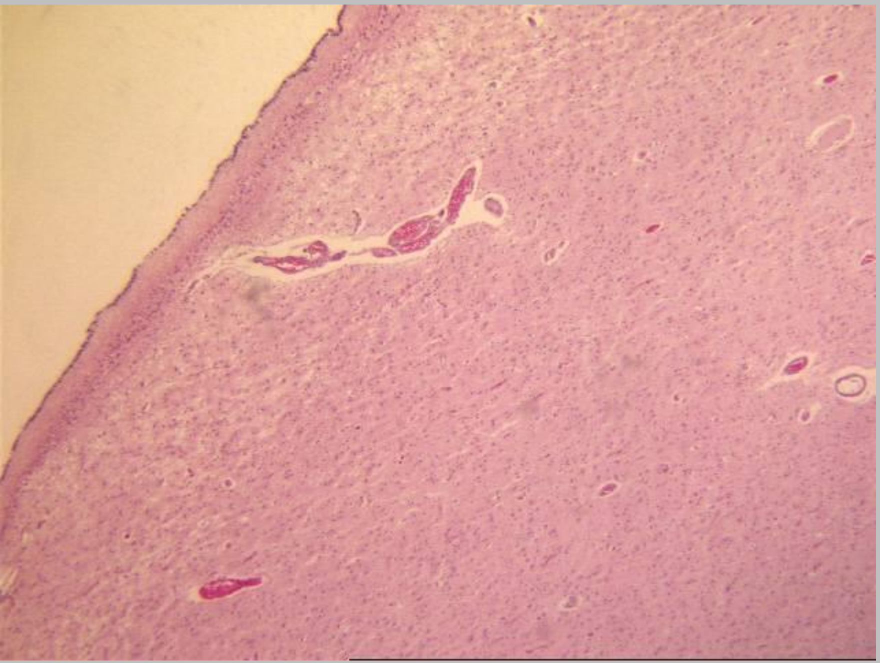
Huntington Disease

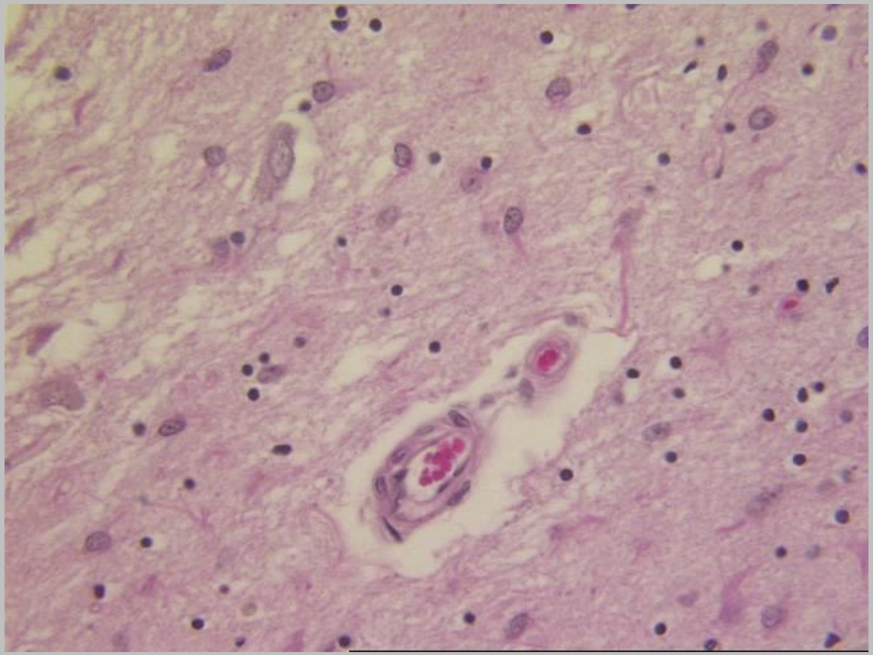


Normal brain (left) v. HD case (right) showing severe atrophy of caudate – notice dilation of ventricles

HD: mild cortical atrophy and severe atrophy of basal ganglia (caudate nucleus >> putamen)

Huntington Disease





Caudate especially shows severe striatal neuronal loss and severe gliosis

.

Direct relationship between degree of degeneration in striatum and severity of clinical symptoms

Protein aggregates containing huntingtin can be found in neurons in the striatum and cerebral cortex

HD – genetics/pathogenesis

- **Prototype of polyglutamine trinucleotide repeat expansion**
- **No sporadic form**
- HD gene, HTT, on chrom. 4 p16.3, encodes **huntingtin protein**
- In first exon of gene - **CAG repeats** which encode polyglutamine region near N terminus of protein; normally 6 to 35 copies of the repeat. If more, HD usually develops - **the longer the repeats, the earlier the onset of disease**
- Repeat expansions occur during spermatogenesis - paternal transmission associated with earlier onset in next generation (known as **anticipation**)
- Expansion of the polyglutamine region = toxic gain-of-function on huntingtin
- Development of intranuclear inclusions containing huntingtin- pathologic hallmark of HD, but not directly involved in cellular injury

Robbins and Cotran, Pathologic Basis of Disease, 10th ed, 2020, Ch. 28

HD

- **Loss of striatal neurons (usually dampen motor activity), causes increased motor output - choreoathetosis**
- Associated cognitive changes probably related to neuronal loss in cerebral cortex
- **Motor symptoms usually precede cognitive impairment**
- Age at onset commonly in fourth and fifth decades - related to length of the CAG repeat in HTT gene
- Movement disorder of HD is **choreiform**, with increased and involuntary jerky movements of all parts of the body; writhing movements of extremities
- Early symptoms of higher cortical dysfunction include forgetfulness, cognitive and affective dysfunction, with progression to severe dementia
- Most common immediate cause of death in HD patients is pneumonia, usually at advanced stage of the disease
- Ethics of screening

Robbins and Cotran, Pathologic Basis of Disease, 10th ed, 2020, Ch. 28

Case Question

- A 50 year old man is brought to his doctor by family members who state that he has become more forgetful in the past few months. He is emotionally labile. His mother dies a few years after experiencing similar symptoms. On physical exam, he has some choreiform movements of his extremities. Which of the following findings is most likely present in this patient?

- A. Decreased levels of hexosaminidase A enzyme
- B. Mutation in presenilin genes
- C. Extra chromosome 21
- D. Abnormal prion protein
- E. Increased trinucleotide CAG repeats

Amyotrophic Lateral Sclerosis

- Progressive disease **- loss of upper motor neurons in cerebral cortex and lower motor neurons in spinal cord and brain stem**
- Neuronal loss causes denervation of muscles – results in **weakness, increasing as disease progresses**
- Incidence about 2/100,000, affects men slightly more than women, usually in 5th decade or later
- Sporadic more common than familial, latter may account for up to 20%
- Familial variant autosomal dominant – mutations in gene encoding copper-zinc **superoxide dismutase (SOD1**) on chrom 21; mutated SOD1 protein misfolds and aggregates, causing cell injury via various mechanisms

- OF INTEREST
- *Most common mutation that gives rise to ALS and FTLD simultaneously - expansion of hexanucleotide repeat in the 5'-untranslated region of a transcript of unknown function, C9orf72 (present in up to 40% of familial ALS and smaller percentage of sporadic cases of ALS)*

Robbins and Cotran, Pathologic Basis of Disease, 10th ed, 2020. Ch. 28.

ALS

- **Asymmetric weakness of hands, early – dropping of objects and difficulty with fine-motor tasks along with cramping and spasticity of arms and legs**

•With progression, decrease in muscle strength and bulk, **fasciculations** (involuntary contractions)

•Eventually affects respiratory muscles, leading to episodes of pneumonia

•Familial cases develop symptoms earlier than sporadic

•Considered to be motor system disease but, significant fraction have more widespread cerebral cortical disease

•Clinical presentation of cerebral disease is usually a frontotemporal dementia, with pathology similar to FTLD associated with TDP-43 inclusions (either behavioral problems or language deficits)

Robbins and Cotran, Pathologic Basis of Disease, 10th ed., 2020, Ch. 28

Amyotrophic Lateral Sclerosis

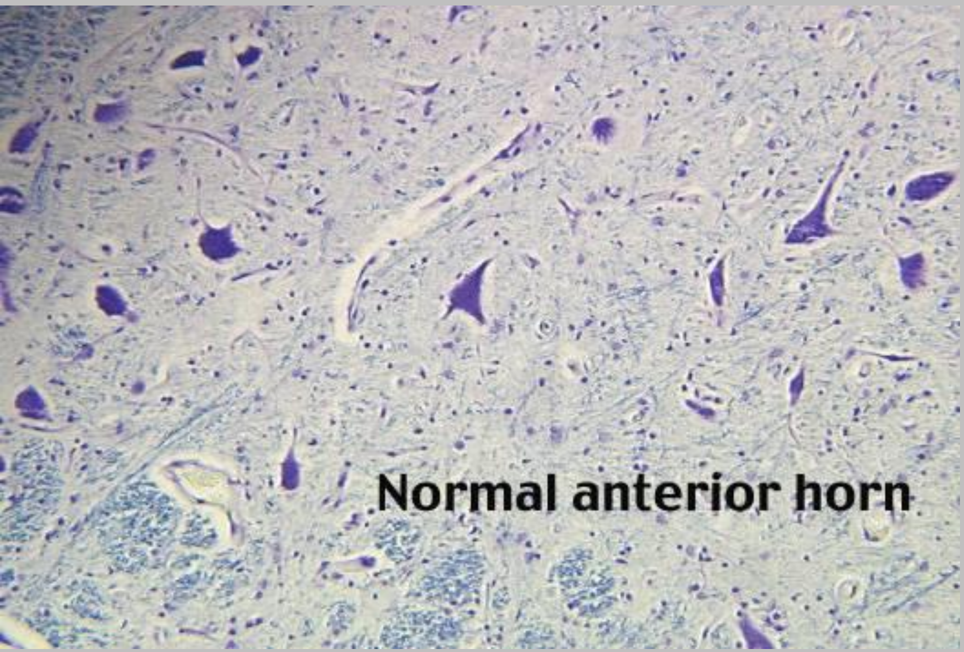
Loss of upper and lower motor neurons.

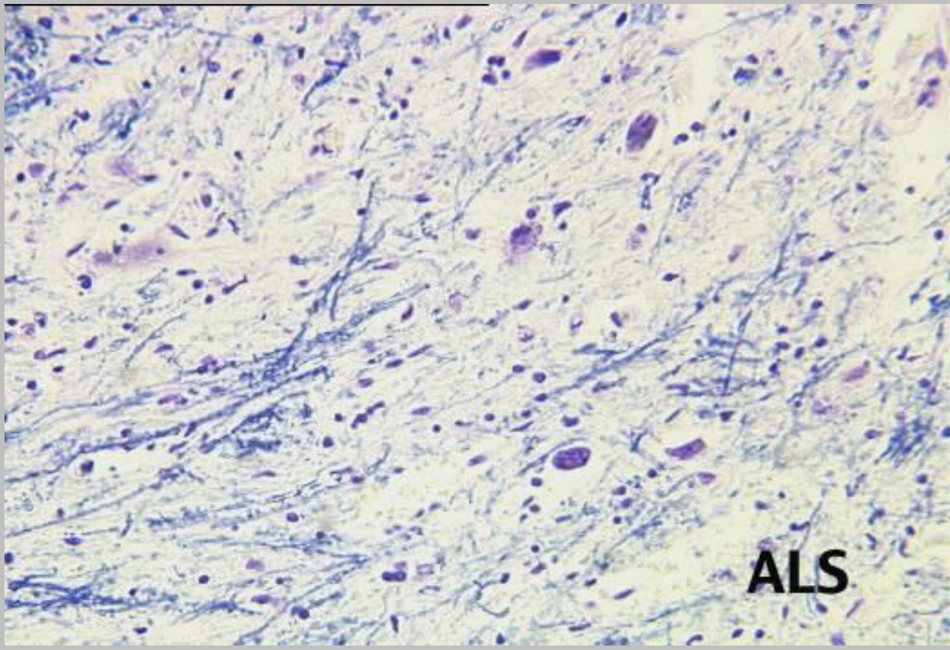
Anterior roots of spinal cord are thin due to loss of lower motor neuron axons

Precentral motor gyrus in cortex may be atrophic in very severe cases.

Robbins and Cotran, Pathologic Basis of Disease, 10th ed., 2020, Ch. 20

Amyotrophic Lateral Sclerosis





.

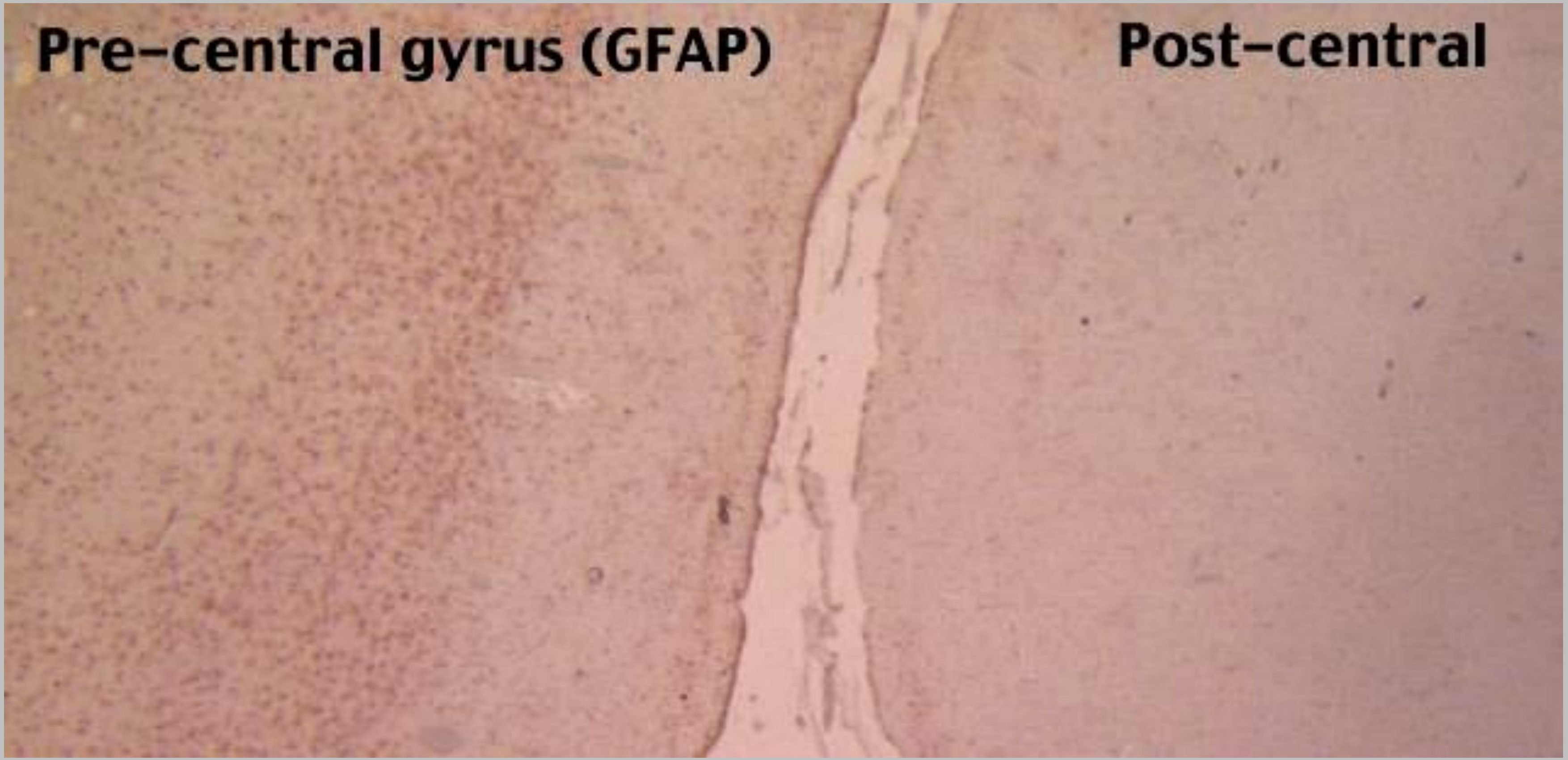
Reduction in number of anterior horn neurons throughout length of spinal cord, associated with reactive gliosis

Skeletal muscles innervated by degenerated lower motor neurons show neurogenic atrophy

Robbins and Cotran, Pathologic Basis of Disease, 10th ed., 2020, Ch. 20

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Amyotrophic Lateral Sclerosis



Loss of neurons in motor cortex is accompanied by gliosis, lacking in the adjacent sensory cortex

Prion Diseases

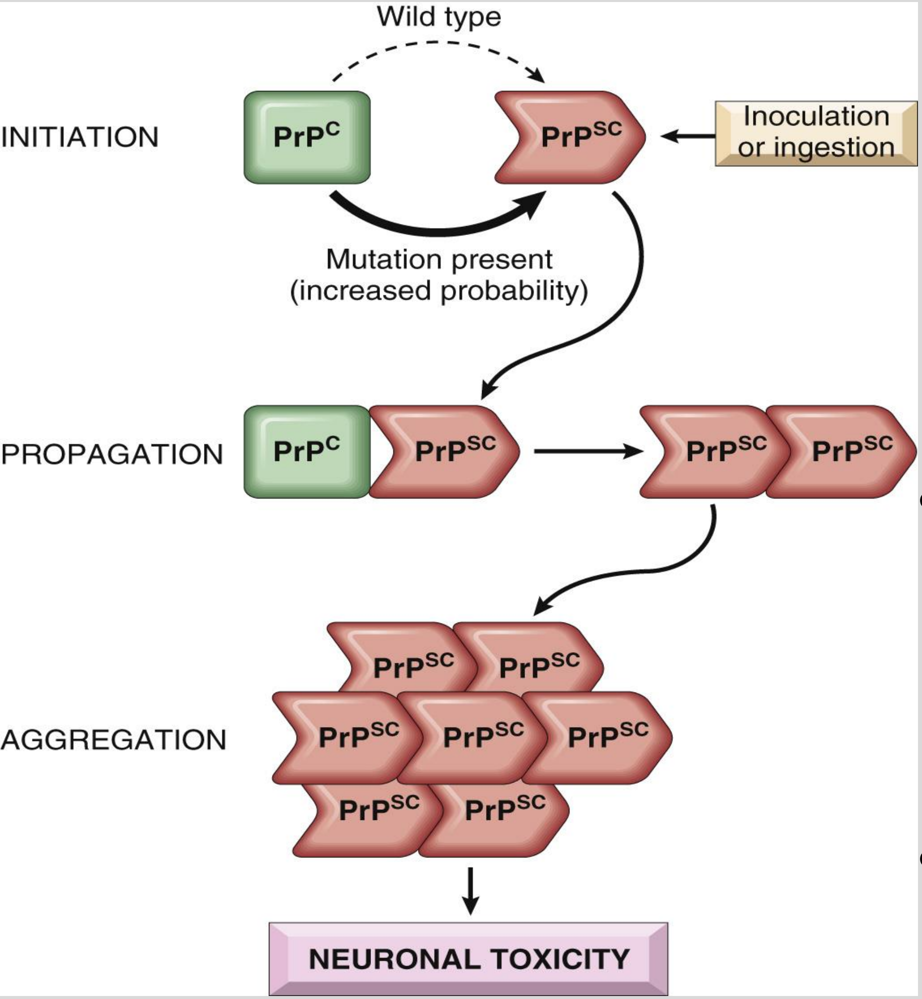
- **Rapidly** progressive neurodegenerative diseases caused by **aggregation and intercellular spread of misfolded (abnormal) prion protein (PrP);** may be **sporadic, familial, transmitted**
- Prion protein present in brain normally; unknown function - Normal secondary structure mainly alpha-helical (PrPc), easily digested by proteases
- Abnormal prion protein (cause of prion diseases) is same protein, secondary structure **mainly beta-pleated sheets (PrPsc) (conformational change)** - more prone to aggregate and resistant to protease digestion, esp. by proteinase K
- Brain grossly shows **"spongiform change"** - intracellular vacuoles in neurons and glia, no inflammation; **clinical rapidly progressive dementia**

Robbins and Cotran, Pathologic Basis of Disease, 10th ed., 2020, Ch. 28.

Pathogenesis

- Conformational change resulting in PrPsc may occur spontaneously at extremely low rate (sporadic CJD) or higher rate if mutations present in PrPc (familial CJD)
- PrPsc facilitates conversion of other PrPc molecules to PrPsc molecules; spreading of PrPsc accounts for **transmissible** variants of prion diseases
- **Abnormal prion protein "recruits" normal prion to abnormal shape (secondary structure),allowing for "propagation"**
- **Allows pathogenic protein to acquire characteristics of infectious agent:** ability to disrupt integrity of normal cellular components through conformational changes

Robbins and Cotran, Pathologic Basis of Disease, 11th ed., 2025, Ch. 28



Robbins and Cotran, Pathologic Basis of Disease, 11th ed., Ch. 20, Fig. 28.34

- α-helical PrPc may spontaneously shift to β-sheet PrPsc conformation (occurs at higher rate in familial diseases associated with germline PrP mutations)
- PrPsc may also be acquired from contaminated food, medical instrumentation, or medicines
- PrPsc converts additional molecules of PrPc into PrPsc through physical interaction, leading to formation of pathogenic PrPsc aggregates

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Case

A 53 year old female with a hx of corneal transplant 6 months ago, presents with loss of balance, mood swings and memory loss. She had not noticed these symptoms until her coworkers and family pointed it out to her. These symptoms presented 4 months ago, and began to interfere with her daily activities. Her ataxia has progressed to the point that she is stumbling and falling.

Her family reports that over the past month her memory quickly deteriorated to the point that she is unable to recognize friends, is unable to drive, is not able to work and forgets if she has eaten. A CT scan of her brain is performed and shows no abnormalities

What abnormal protein in the brain is associated with this condition?

A. Alpha-synuclein
B. Beta-amyloid
C. Tau
D. PrPsc
E. Myelin

Creutzfeldt-Jakob Disease

- **Most common prion disease**, rare disorder - rapidly progressive dementia
- Sporadic form of CJD - annual incidence approximately 1 per 1,000,000; accounts for about 90% of CJD cases
- Familial forms caused by mutations in PRNP, gene that encodes PrP
- May also have iatrogenic transmission, from corneal or dural transplantation, deep implantation of electrodes in brain, and administration of contaminated cadaveric human growth hormone
- Starts with subtle changes in memory and behavior followed by **rapidly progressive dementia**
- Often associated with pronounced involuntary jerking muscle contractions on sudden stimulation (startle myoclonus); ataxia, (cerebellar involvement) present in minority
- **Uniformly fatal; average survival 7 months after onset of symptoms**

Robbins and Cotran, Pathologic Basis of Disease, 11th ed., 2025, Ch. 28

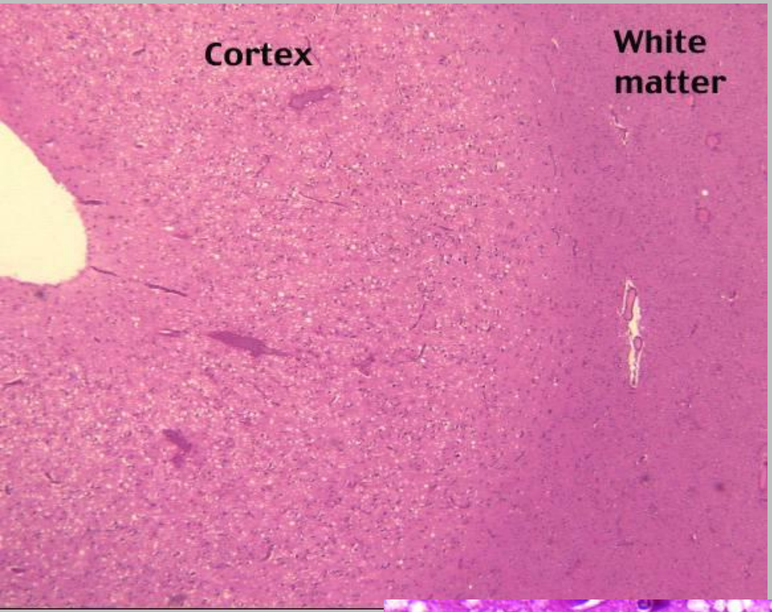
CJD

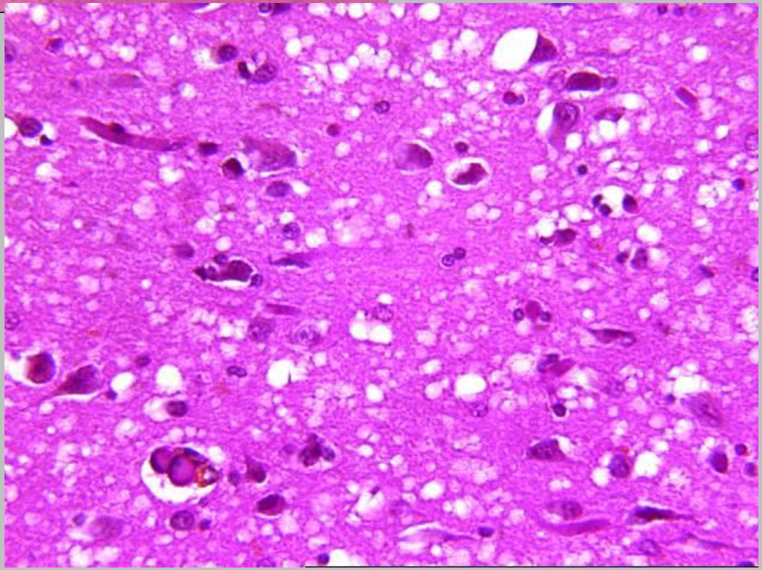
- Most cases show **little or no cerebral atrophy** – clinically rapid, progressive dementia
- **Fatal within 7 - 12 months of onset.**
- **90% cases sporadic**
- Familial forms caused by mutations in PRNP

Robbins and Cotran, Pathologic Basis of Disease, 11th ed., 2025, Ch. 28

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

CJD



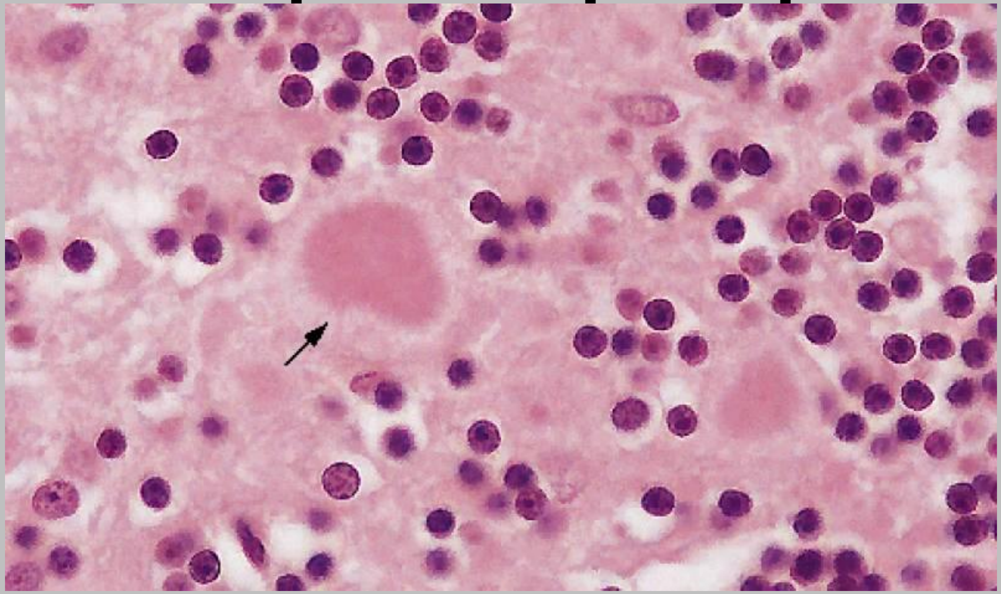


- **Spongiform changes** of both cortical and deep gray matter, neuronal loss and gliosis
- Uneven formation of small and empty microscopic vacuoles of varying sizes within neuropil and sometimes in perikaryon of neurons - **inflammation absent**

Robbins and Cotran, Pathologic Basis of Disease, 10th ed., 2020, Ch. 28

CJD - prion plaques

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

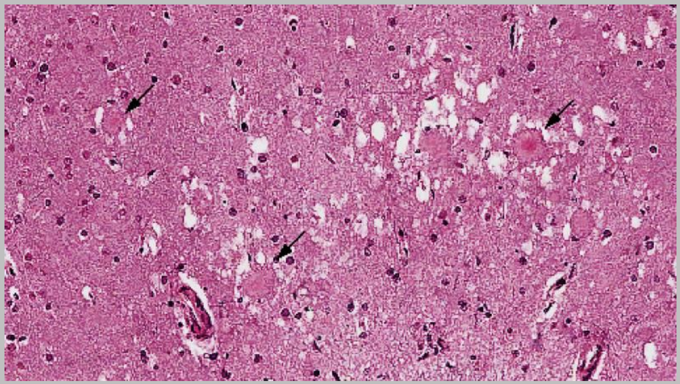


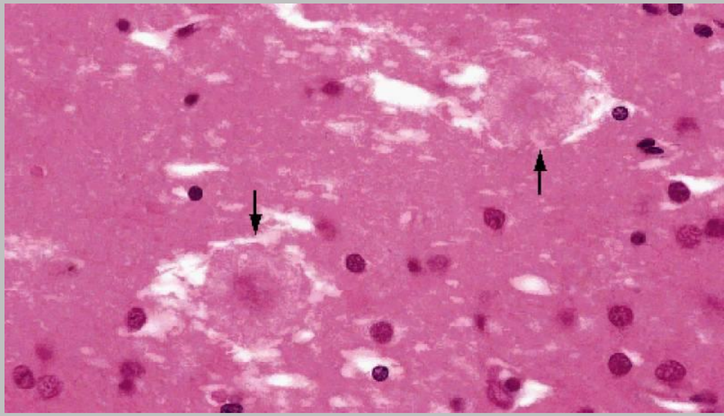
- In 15% of CJD cases “kuru” (prion) plaques found in cerebellum: extracellular deposits of aggregated abnormal protein: Congo red + and PAS +
- Abundant in cerebral cortex in vCJD – see next slide
- IHC staining shows presence of proteinase K–resistant PrPsc in tissue (all forms of prion disease)
- Note- Kuru is **transmissible** neurodegenerative disease first seen in Papua New Guinea (funerary cannibalism) – kuru means “trembling” - tremors and loss of coordination from neurodegeneration

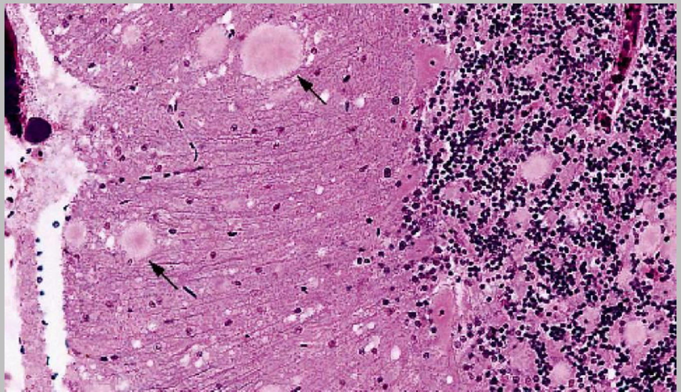
Robbins and Cotran, Pathologic Basis of Disease, 10th ed., 2020, Ch. 28

Variant CJD

- **1995 UK – new CJD-like illness**
- Clinical- young adults, behavior disorders prominent early and neuro more slowly progressive.
- Linked to consumption of bovine spongiform encephalopathy agent from eating contaminated foods or by transfusion of blood from patients in asymptomatic/preclinical stage of vCJD.
- Public health measures limited spread of vCJD
- Incidence in UK peaked in 2000
- vCJD = cortical and cerebellar "florid" plaques and moderate spongiform change







Robbins and Cotran, Pathologic Basis of Disease, 11th ed., 2025, Ch. 28

SELECTED NEUROPATHIES

- Charcot-Marie Tooth disease
- Diabetic neuropathy
- Traumatic/compression neuropathy
- (Metabolic/nutritional/hormonal peripheral neuropathies - Renal failure, Thiamine deficiency, Vitamin B12 deficiency, Vitamin B6 deficiency, thyroid dysfunction (compression), deficiencies in copper and zinc)
- (Toxic – alcohol, lead, mercury, arsenic, thalium, chemotherapy)
- (Neuropathies associated with malignancy- Direct compression/infiltration by tumor, paraneoplastic effects)

Hereditary Motor and Sensory Neuropathies Charcot-Marie Tooth Disease

Most common inherited peripheral neuropathies, affecting up to 1 in 2500; clinical phenotype can result from mutations in more than 50 different genes

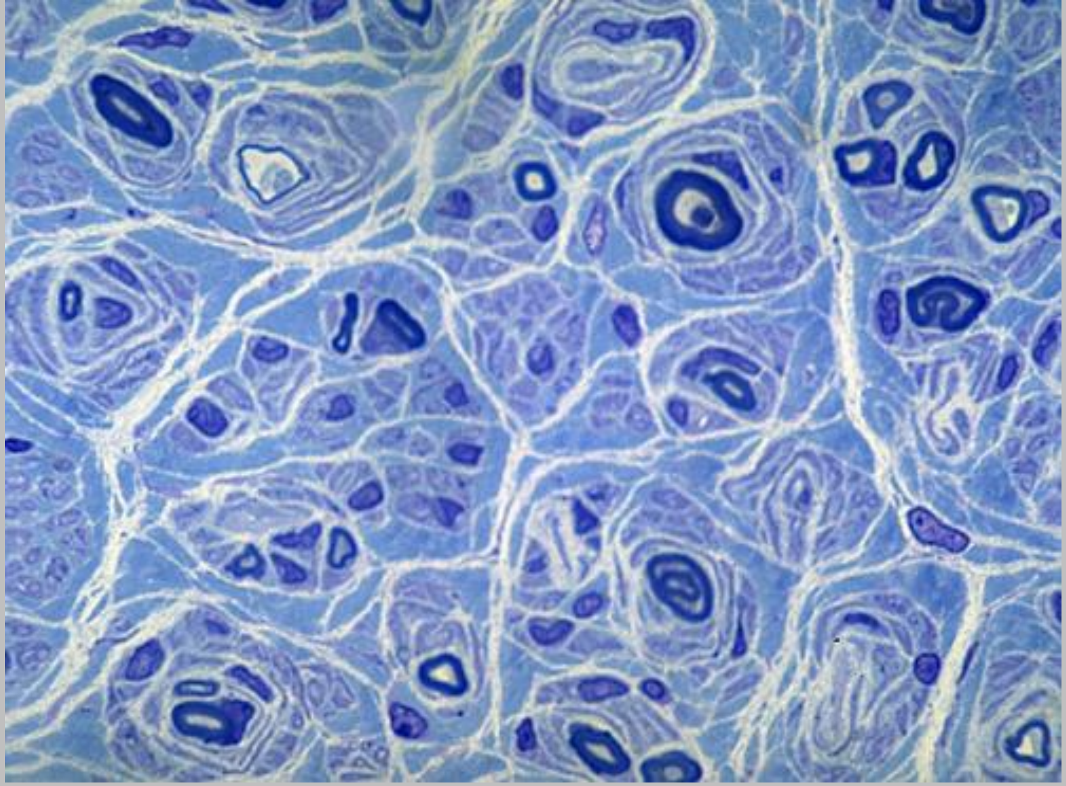
- **CMT1 - group of AD demyelinating neuropathies, collectively most common subtype of hereditary motor and sensory neuropathy**
  - CMT1A: 55% of genetically defined CMT - caused by duplication of region on chromosome 17 that includes peripheral myelin protein 22 (PMP22) gene; usually presents in second decade of life - slowly progressive distal demyelinating motor and sensory neuropathy
  - CMT1B: caused by mutations in the myelin protein zero gene (MPZ) and accounts for about 9% of genetically defined cases of CMT
  - Usually presents in childhood/early adulthood, normal life span
  - **Peroneal muscular atrophy**
  - **Distal muscle weakness, atrophy of leg beneath knee, secondary orthopedic issues (pes cavus)**
- OF INTEREST:
- *CMTX - X-linked forms of CMT disease - mutations in GJB1 gene, encodes connexin32, gap junction component expressed in Schwann cells.*
- *CMT2 includes autosomal dominant neuropathies associated with axonal injury (not demyelinating); caused by mutations in MFN2 gene, needed for normal mitochondrial fusion - Usually severe, disease onset in early childhood*

Robbins and Cotran, Pathologic Basis of Disease, 10th ed., 2020, Ch. 27

Charcot-Marie-Tooth disease
Classification & external resources

The foot of a person with Charcot-Marie-Tooth. The lack of muscle, a high arch, and hammer toes are signs of the genetic disease.

progressive muscular atrophy of the calf

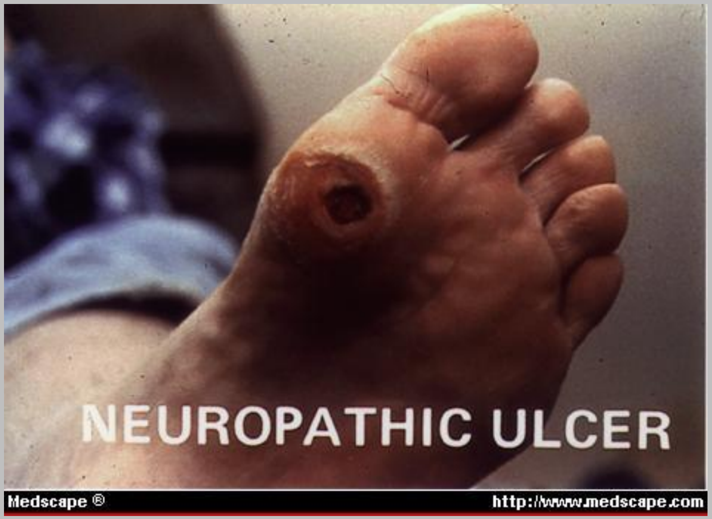


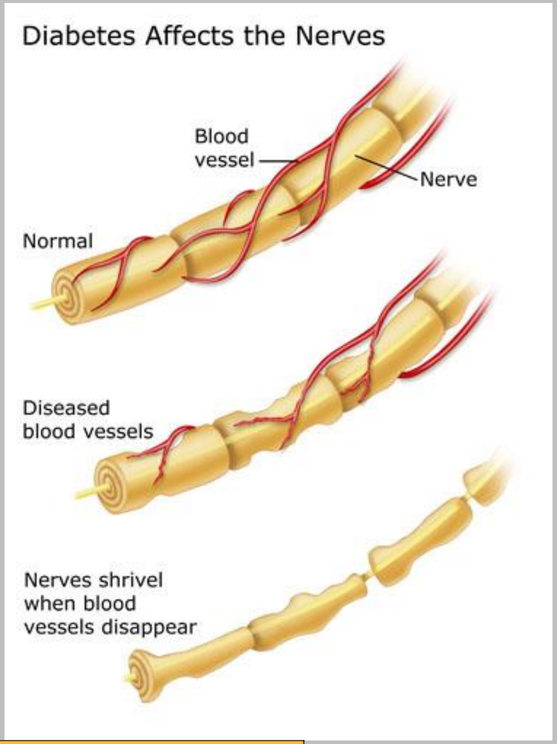
CMT - Hypertrophic changes: **"Onion bulb" formations**: concentric layers of Schwann cell processes and collagen around axons, caused by repetitive segmental demyelination and regeneration of myelin. May cause gross thickening of peripheral nerves (**hypertrophic neuropathy**) which can be clinically palpated.

Peripheral Neuropathy - Diabetes

- **DM – most common peripheral neuropathty** - up to 50% with diabetes overall and up to 80% who have had DM for more than 15 years
- **Sensimotor (distal symmetric sensory), autonomic, and/or focal or multifocal asymmetric**
- **Sensimotor (most common) – ascending distal symmetric sensorimotor polyneuropathy**
  - Major finding - axonal neuropathy with some segmental demyelination
  - Endoneurial arterioles show thickening, hyalinization, and extensive reduplication of basement membranes
  - Usually presents with sensory symptoms - numbness, loss of pain sensation, difficulty with balance, and paresthesias or dysesthesias
  - **“stocking and glove” – decreased sensation in distal extremities; loss of pain can lead to ulcers which heal poorly because of vascular injury in diabetes (may require amputation)**
- **Dysfunction of autonomic system** affects 20% - 40% of individuals with DM, in association with a distal sensorimotor neuropathy - includes, **postural hypotension, incomplete emptying of the bladder (resulting in recurrent infections), and erectile dysfunction.**

Robbins and Cotran, Pathologic Basis of Disease, 10th ed., 2020, Ch. 27





- Mechanism of diabetic neuropathy complex, not completely resolved
- Both **metabolic and secondary vascular changes** may contribute to damage of axons and Schwann cells

- Hyperglycemia causes nonenzymatic glycosylation of proteins, lipids, and nucleic acids - resulting in advanced glycosylation end-products (AGEs) - may interfere with normal protein function and activate inflammatory signaling
- Excess glucose in cells reduced to sorbitol - depletes NADPH and increases intracellular osmolality
- May predispose peripheral nerves to injury by reactive oxygen species

Robbins and Cotran, Pathologic Basis of Disease, 10th ed., 2020, Ch. 27

Traumatic neuropathy/Compression neuropathy

- **Trauma** - Axonal regeneration slow - regrowth complicated by **discontinuity between proximal and distal parts of nerve sheath/misalignment of nerve fascicles**
  - If axons continue to grow, (distal segments not positioned correctly)- **mass of tangled axonal processes result: Traumatic or amputation neuroma;** histo **-** small bundles of axons randomly oriented, each surrounded by organized layers of Schwann cells, fibroblasts, and perineural cell

- **Compression** – When peripheral nerve has chronic increased pressure
  - **Carpal tunnel syndrome**, most common entrapment neuropathy - **compression of median nerve at level of wrist within compartment delimited by transverse carpal ligament**
  - Women more commonly affected than men, frequently bilateral
  - May be associated with tissue edema, pregnancy, inflammatory arthritis, hypothyroidism, amyloidosis, acromegaly, diabetes mellitus, and excessive repetitive motions of the wrist
  - Symptoms limited to dysfunction of median nerve - **include numbness and paresthesias of tips of thumb and first two digits**

Robbins and Cotran, Pathologic Basis of Disease, 10th ed., 2020, Ch. 27

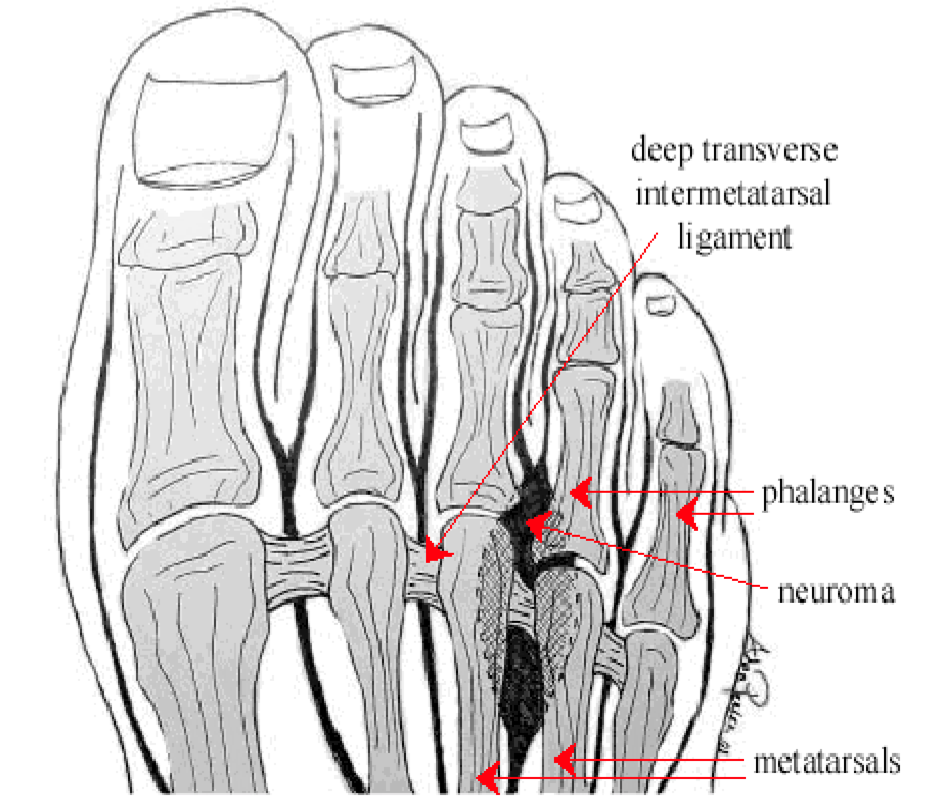
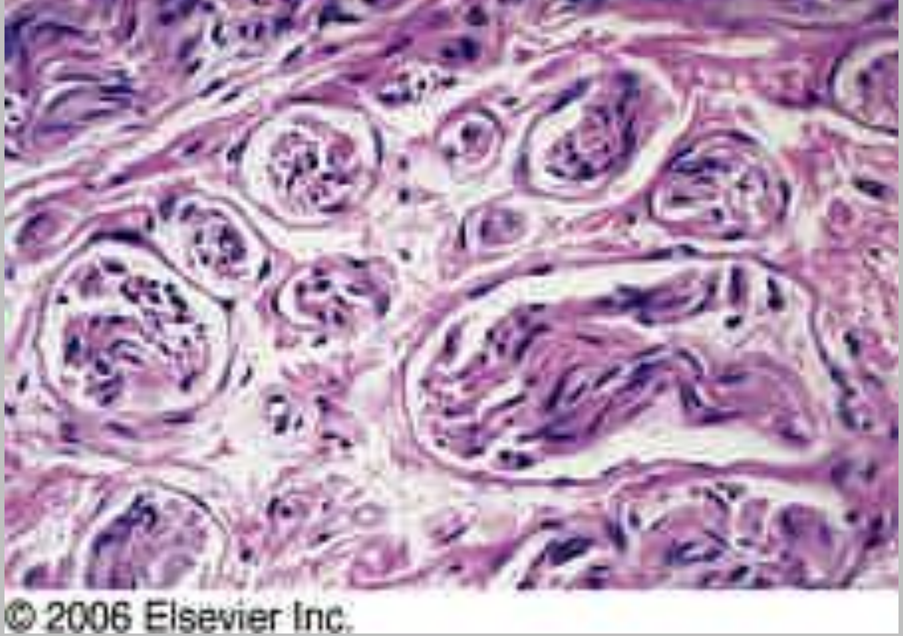


Figure 1. Morton's neuroma found between 3rd and 4th toes underneath the deep transverse intermetatarsal ligament.



- Morton neuroma – compression neuropathy in foot, affecting interdigital nerve at intermetatarsal sites, histo – perineural fibrosis
- Occurs more often in women than in men (high heels)
- Leads to foot pain (metatarsalgia)

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Robbins and Cotran, Pathologic Basis of Disease, 10th ed., 2020, Ch. 27

Italo Maratta

January 5, 1934 – March 8, 2021

Summary Slide

- Alzheimer's disease – dementia, global atrophy, hydrocephalus ex vacuo, **neuritic plaques** contain **amyloid**, **neurofibrillary tangles** have **tau protein**
- Pick's disease (FTLD-tau) – dementia, similar to Alzheimer's disease, frontotemporal, knife-edge gyri
- Parkinson's disease – loss of dopamine in substantia nigra, movement disorder, **Lewy bodies** contain **alpha-synuclein**, tremor, bradykinesia
- Huntington's disease – AD, movement disorder which can also have dementia, atrophy of caudate**, CAG repeats** (greater number earlier onset of disease), chorea
- ALS – upper and lower motor neuron degeneration
- Prion encephalopathies – CJD – abnormal configuration of prion proteins, rapid dementia, sporadic, hereditary, and acquired forms, **Kuru plaques**
- Charcot-Marie-Tooth disease – Hereditary motor and sensory neuropathy: distal muscle atrophy (peroneal), sensory loss, foot deformities (pes cavus), segmental demyelination, palpable peripheral nerves
- Diabetic, traumatic, and compression neuropathies

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Lecture Feedback Form:

https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R